

Writeup

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1 Introduction

Quantum networks can be represented using graph states [5], and the qubit-minor problem is to determine whether it is possible to obtain a certain graph state using local Clifford operations, Pauli measurements, and classical communication on a starting graph state [3]. It is desirable to establish maximally entangled pairs of qubits to enable quantum communication [1].

Ref. [3] proved that the qubit-minor problem is equivalent to the vertex-minor problem. Ref. [4] investigated extracting 2 Bell pairs simultaneously and proved that bottlenecks cannot be avoided in some cases in line and ring graph networks. This was furthered by Ref. [8], which solved the Bell vertex-minor problem for line, ring, and tree graphs. Finally, Ref. [7] defined the X protocol for extracting a Bell pair and presented cases in the 2×3 and 2×4 grids where bottlenecks cannot be overcome.

In Section 2, we provide graph theory definitions, auxiliary protocols for later constructions, and statements of necessary theorems from Ref. [2], [3], [7], and [8]. In Section 3, we provide proofs of bottlenecks and constructions for extracting various Bell pairs from the $2 \times n$ grid. In Section 4, we give constructions for the $3 \times n$ grid. Finally, we completely solve the tadpole graph, then define the double ring graph and solve some cases on this graph in Section 5.

2 Preliminaries

Definition 2.1. Given $|V|$ qubits and a set of edges E , the graph state is defined to be

$$|G\rangle = \prod_{(i,j) \in E} CZ_{ij} |+\rangle^{\otimes V}, \quad (1)$$

where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

Definition 2.2. Let $G = (V, E)$ be a graph, and let $v \in V$. The neighborhood of v is defined as follows:

$$N_v = \{u \in V : \{v, u\} \in E\}. \quad (2)$$

Definition 2.3. The adjacency matrix A of a graph $G = (V, E)$ is the $|V| \times |V|$ matrix satisfying

$$A_{ij} = \begin{cases} 0 & \{i, j\} \notin E \\ 1 & \{i, j\} \in E \end{cases} \quad (3)$$

Definition 2.4. Let $G = (V, E)$ be a graph, and let $v \in V$. $G \setminus v$ is the graph on vertices $V \setminus v$ and edges $E \setminus \{\{v, u\} : u \in N_v\}$.

Definition 2.5. Let $G = (V, E)$ be a graph with adjacency matrix A , and let $v \in V$. $\tau_v(G)$, the graph obtained by performing a local complementation with respect to v on G , is the graph with adjacency matrix

$$A + B \pmod{2}, \quad (4)$$

where B is the adjacency matrix of the complete graph on the vertices of N_v . If a graph G' can be obtained by performing a sequence of local complementations on G , then G and G' are LC-equivalent ($G \sim_{\text{LC}} G'$).

Definition 2.6. Let $G = (V, E)$ be a graph, and let $v \in V$. v is a leaf if $\exists u \in V$ such that $N_v = \{u\}$. We say v is a leaf of u .

v is an axil if $\exists u \in V$ such that $N_u = \{v\}$. We say v is the axil of u .

v is a twin if $\exists u \in V$ such that $u \neq v$ and $N_v \setminus \{u\} = N_u \setminus \{v\}$. We say v is a twin of u .

If v is a leaf, axil, or twin of u , then v and u are foliage equivalent ($v \sim_{\text{F}} u$).

Definition 2.7. Let $G = (V, E)$ and $G' = (V', E')$ be graphs. G' is a vertex-minor of G ($G' < G$) if one can obtain G' by performing a sequence of vertex deletions and local complementations on G .

Lemma 2.1. (Proposition II.1 of [8]) Let G, G' , and G'' be graphs. If $G' < G$ and $G'' < G'$, then $G'' < G$.

Definition 2.8. Let $G = (V, E)$ be a graph, and let $v \in V$. $Z_v(G) := G \setminus v$ is the graph obtained by performing a Z-measurement of v . $Y_v(G) := Z_v \circ \tau_v(G)$ is the graph obtained by performing a Y-measurement of v . Let $u \in N_v$. $X_v(G) := Z_v \circ \tau_u \circ \tau_v \circ \tau_u(G)$ is the graph obtained by performing an X-measurement of v .

Theorem 2.1. (Theorem 3.2 in [3]) Let $G = (V, E)$ and $G' = (V', E')$ be graphs and $U = (v_1, \dots, v_{n-k})$ be the any ordering of the set $V \setminus V'$. Let $\mathcal{P}_U$ denote the set of graph operations

$$\mathcal{P}_U = \{P_{v_{n-k}} \circ \dots \circ P_{v_1} : P_v \in \{X_v, Y_v, Z_v\}\}. \quad (5)$$

Then

$$G' < G \iff \exists P \in \mathcal{P}_U : G' \sim_{\text{LC}} P(G). \quad (6)$$

Theorem 2.2. (Source Reduction; Theorem 2.7 in [2]) Let G and G' be graphs such that G' contains no isolated vertices. Let v be a vertex in G but not in G' . Then

- if v is a leaf or twin, then $G' < G \iff G' < (G \setminus v)$
- if v is an axil, then $G' < G \iff G' < (\tau_w \circ \tau_v(G) \setminus v)$, where w is a leaf associated to v

Lemma 2.2. (Lemma IV.1 in [8]) Given graphs $G' < G$ with a set of pairwise foliage-equivalent vertices $A \subset G$, the vertices in $A \cap G'$ are either pairwise foliage-equivalent to each other, all isolated vertices, or the empty set.

Theorem 2.3. (Target Reduction; Theorem IV.2 in [8]) Let G and G' be graphs with $G' < G$ and vertices $v \sim_F w$ in both G and G' .

- If v is a leaf of w or v is a twin of w in G , let $\tilde{G} = G \setminus v$
- If w is a leaf of v in G , let $\tilde{G} = \tau_w \circ \tau_v(G) \setminus v$
- If v is a leaf of w or v is a twin of w in G' , let $\tilde{G}' = G' \setminus v$
- If w is a leaf of v in G , let $\tilde{G}' = \tau_w \circ \tau_v(G') \setminus v$

Theorem 2.4. (Theorem V.1 in [8]) Two Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ can be extracted from L_n using local Clifford operations, Pauli measurements, and classical communication iff $a_1 < a_2 < b_1 < b_2$ (up to some reordering) and a_2 and b_1 are not adjacent.

Lemma 2.3. (Lemma V.1 in [8]) It is not possible to extract a Bell pair $\{a_1, a_2\}$ and an isolated, unmeasured qubit b from R_n using local Clifford operations, Pauli measurements, and classical communication if a_1, a_2, b are consecutive, in that order.

Theorem 2.5. (Theorem V.3 in [8]) Two Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ can be extracted from R_n using local Clifford operations, Pauli measurements, and classical communication iff $a_1 < a_2 < b_1 < b_2$ (up to some reordering) and no three of $\{a_1, a_2, b_1, b_2\}$ are consecutive.

Throughout, we number the vertices of an $m \times n$ grid such that the vertex in row x and column y is labeled $nx + y$.

Theorem 2.6. (from Section IV of [7]) Let G be a $2 \times n$ grid. If $n > 2$ is odd, then it is possible to extract Bell pairs $\{1, 2n\}$ and $\{n, n+1\}$. If n is even, then it is possible to extract Bell pairs $\{1, n\}$ and $\{n+1, 2n\}$.

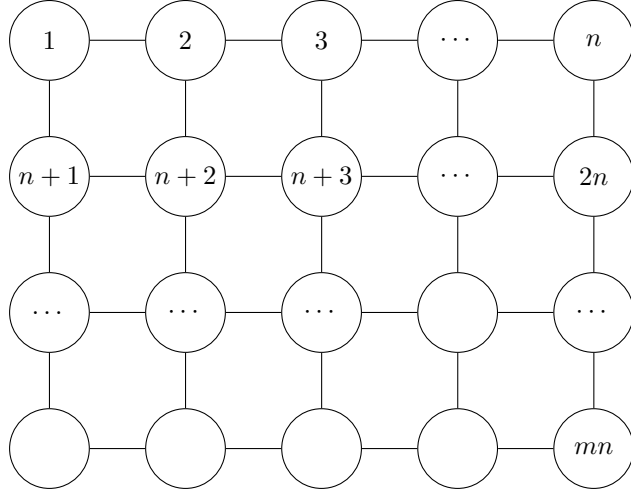


Figure 1: The $m \times n$ grid

Definition 2.9. Let $G = (V, E)$ be a graph. A set $X \subseteq V$ is a **local set** iff there exists some $S \subseteq V$ such that $X = S \cup \text{Odd}_G(S)$, where

$$\text{Odd}_G(S) = \{v \in V \mid |N(v) \cap S| \text{ is odd}\}.$$

We will call S a **generator set** for X .

Theorem 2.7 (Lemma 3 in [6]). Local complementation preserves local sets. In other words, given a local set S of $G = (V, E)$, for any $v \in V$, S is a local set of $\tau_v(G)$.

Corollary 2.1. Let $G = (V, E)$ be a graph and $G' = (V', E')$ be a vertex minor of G . If $S \subseteq V'$ is a local set of G , S is a local set of G' .

2.1 Auxiliary Protocols

In this subsection, we define several auxiliary protocols that will be used in the proofs of our constructions for grid graphs.

Definition 2.10. Let $G = (V, N)$ be a $3 \times n$ grid. Suppose $n \geq 3$ and let vertex $a = n + x$ for some $x \in \mathbb{N}$ s.t $1 < x < n$, see figure Figure 2. Define the **Crossing Protocol** to be

$$\chi_a(G) = [\tau_x \circ \tau_a \circ \tau_x(G)] \setminus a \setminus x \setminus (2n + x) \quad (7)$$

If $n \geq 3$, it can be shown that the resulting graph will look like figure Figure 2, if we choose $a = n + x$ s.t $1 < x < n$. See appendix A.1

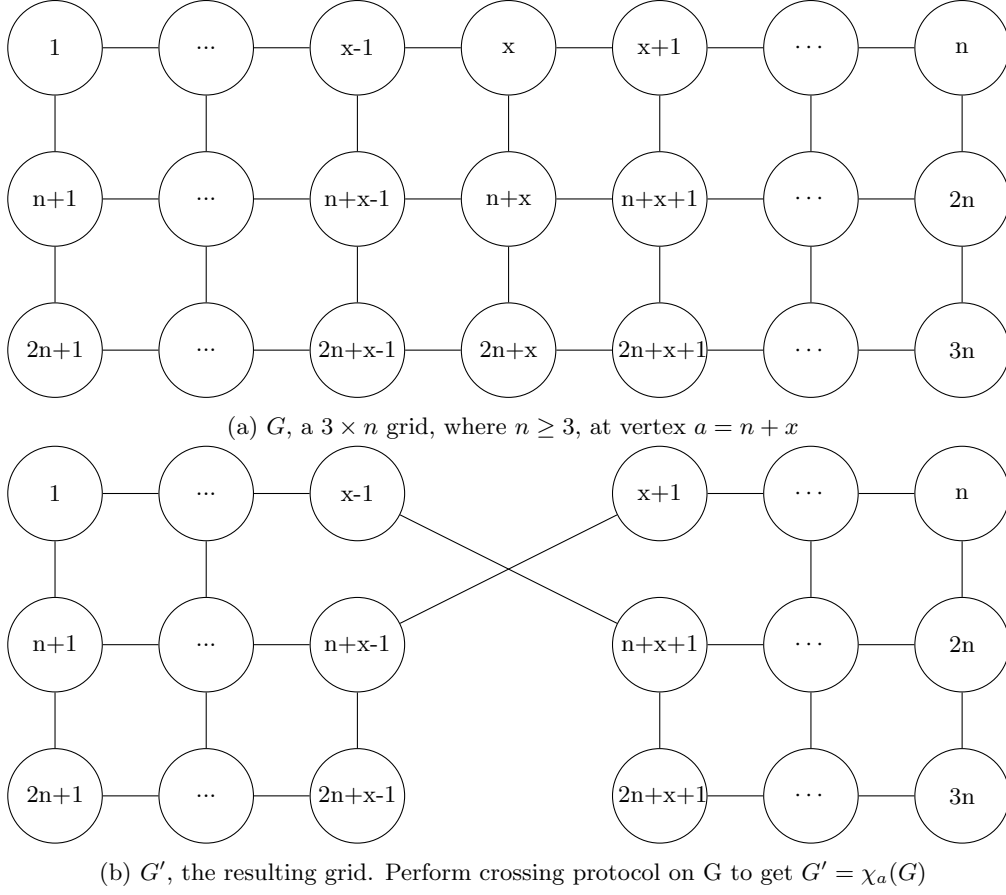


Figure 2

Definition 2.11. Let $G = (V, N)$ be a $3 \times n$ grid. Suppose $n \geq 4$. Define the **Flipping Protocol** to be

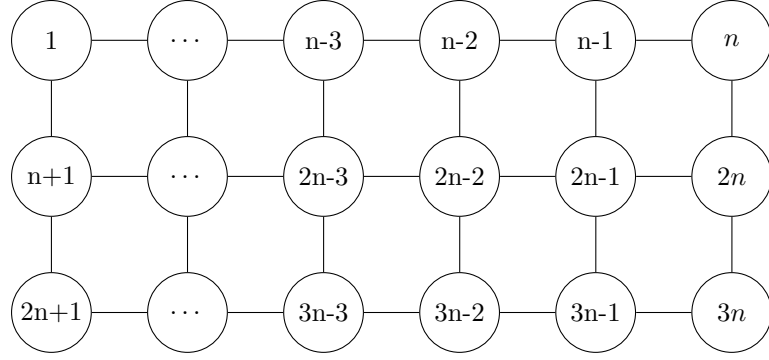
$$\varphi(G) = Y_{3n-2} \circ Y_{2n-2} \circ Y_{n-2} \circ \tau_{2n} \circ \tau_{3n} \circ \tau_{2n} \circ Z_{3n-1} \circ X_{2n-1}^{3n-1} \circ X_{n-1}^n(G) \quad (8)$$

If $n \geq 4$, $G' < G$ where G' is a $3 \times (n-2)$ grid. It can be shown that the resulting graph, G' , will look like G , with the $n-1$ column deleted and the n column replacing the $n-2$ column. Additionally, the corner vertices, $\{n\}$ and $\{3n\}$ will have swapped places. See fig Figure 3. See appendix A.2.

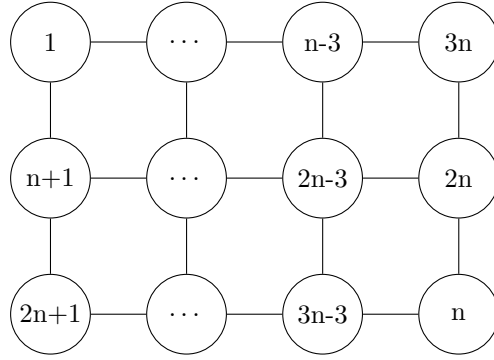
Definition 2.12. Let $G = (V, N)$ be a $3 \times n$ grid, where $n \geq 3$. Define the **Corner Extraction Protocol** to be

if $n = 3$

$$\kappa(G) = Y_9 \circ Y_6 \circ Z_8 \circ X_5^8 \circ Y_2(G) \quad (9)$$



(a) G , a $3 \times n$ grid, where $n \geq 3$



(b) G' , the resulting grid. Perform flipping protocol on G to get $G' = \varphi(G)$

Figure 3

if $n \geq 4$

$$\kappa(G) = Z_{2n+4} \circ Z_{n+4} \circ Y_{2n+3} \circ Y_{n+3} \circ Z_{2n+2} \circ X_{n+2}^{2n+2} \circ Y_2(G) \quad (10)$$

If $n = 3$, $G' < G$ where G' contains the extracted bell pairs $\{1, 7\}$ and $\{4, 3\}$. If $n \geq 4$, $G' < G$ where G' is a $3 \times (n - 4)$ grid containing all remaining columns greater than 4. Additionally, the vertices $\{1, 2n + 1\}$ have been extracted and the vertices $\{n + 1, 3, 4\}$ form a line graph attached to the top left corner of the $3 \times (n - 4)$ grid. See fig Figure 4. See appendix A.3.

Definition 2.13. Let $G = (V, N)$ be a $2 \times n$ grid, where $n \geq 3$. Let vertex c be the corner vertex, $c = n + 1$, and vertex t s.t $c < t < 2n$. Define the **Corner Replacement Protocol** to be

$$\rho_c^t(G) = \tau_c \circ Y_t \circ \cdots \circ Y_{c+2} \circ Y_{c+1} \circ Z_{t-n-1} \circ \cdots \circ Z_2 \circ Z_1(G) \quad (11)$$

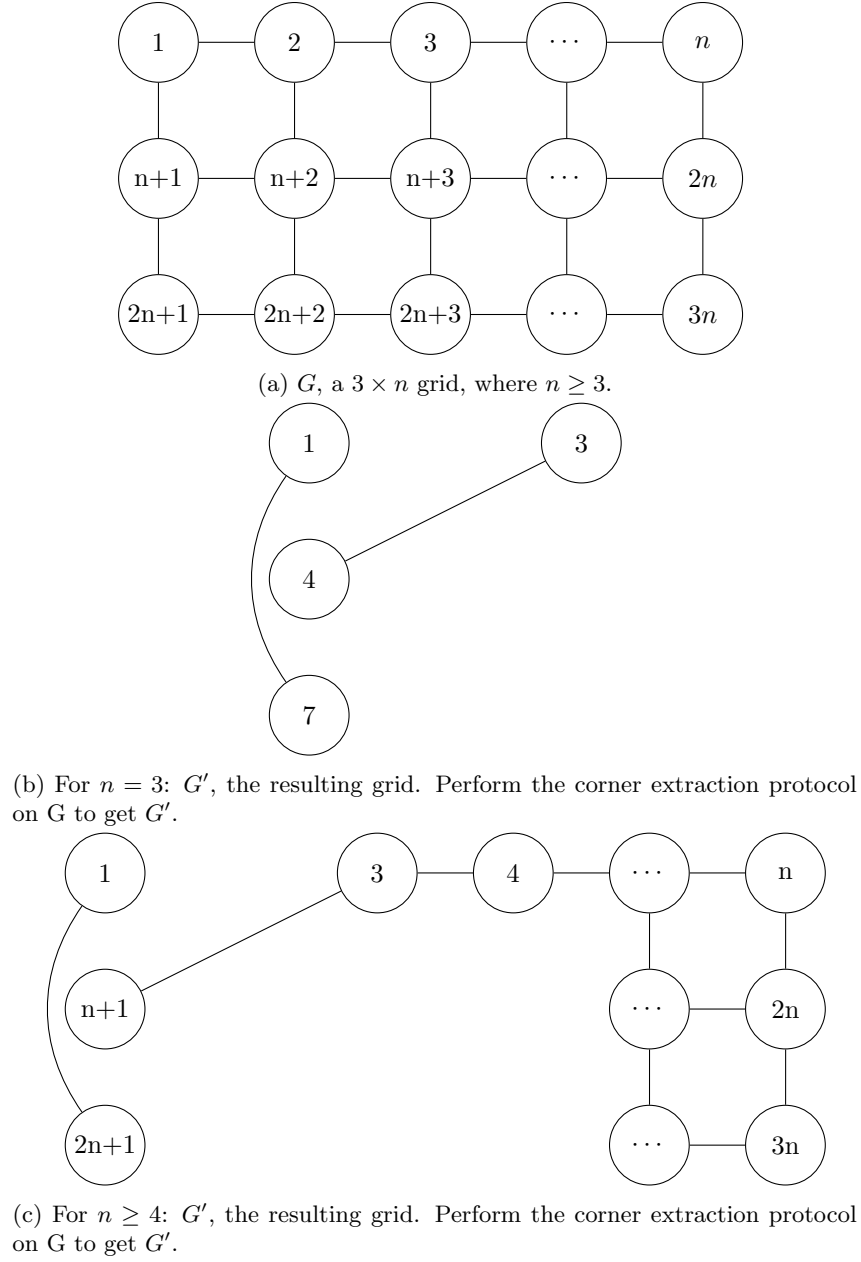
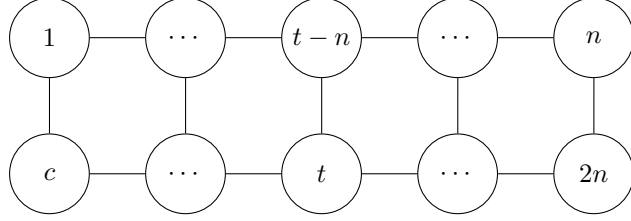


Figure 4

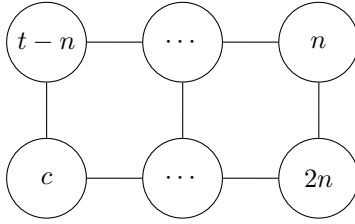
It can be shown that $G' < G$ where G' is a $2 \times (n - (t - n) + 1)$ grid with corner vertices $V_c \in \{t - n, c, n, 2n\}$. See fig Figure 5. See appendix A.4.

*Note: If vertex c is a right corner or a top corner, this protocol can be applied by inverting the vertices the operations are done on horizontally (if right side) or vertically (if

top row).



(a) G is a $2 \times n$ grid, where $n \geq 3$. Vertex $c = n + 1$, and vertex t is s.t $c < t < 2n$.



(b) G' , the resulting grid. G' is a $2 \times (n - (t - n) + 1)$ grid with corner vertices $V_c \in \{t - n, c, n, 2n\}$.

Figure 5

Definition 2.14. Let $G = (V, N)$ be a $2 \times n$ grid, where $n \geq 5$. Define the **Corner Folding Protocol** to be

$$\xi(G) = \tau_{2n} \circ \tau_{n+1} \circ Y_{n-1} \circ Y_2 \circ Y_n \circ Y_1 \circ \tau_{n-1} \circ \tau_2 \circ \tau_n \circ \tau_1 \circ \tau_{2n} \circ \tau_{n+1}(G) \quad (12)$$

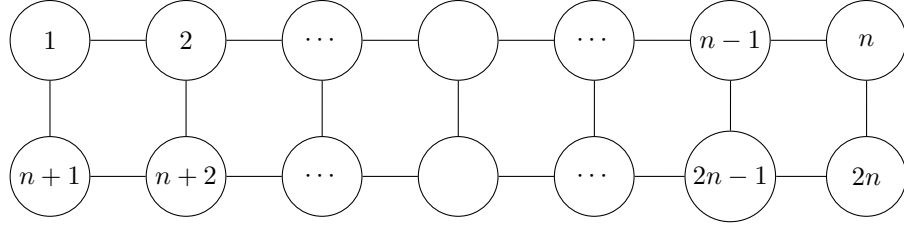
It can be shown that $G' < G$ where G' is a $2 \times (n - 2)$ grid with corner vertices $V_c \in \{n + 1, n + 2, 2n, 2n - 1\}$. See fig Figure 6. See appendix A.5.

*Note: If you want to 'fold over' the top corners, a similar protocol can be used by vertically inverting the vertices the operations are being performed on.

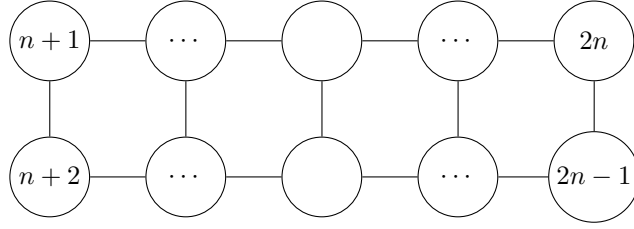
3 $2 \times n$ Grids

3.1 Corner Vertices

Lemma 3.1. Let G be a $2 \times n$ grid, where $n > 4$. If n is odd, then one can extract Bell pairs $\{1, n\}$ and $\{n + 1, 2n\}$ using local Clifford operations, Pauli measurements, and classical communication. If n is even, one can extract Bell pairs $\{n, n + 1\}$ and $\{1, 2n\}$.



(a) G is a $2 \times n$ grid, where $n \geq 5$.



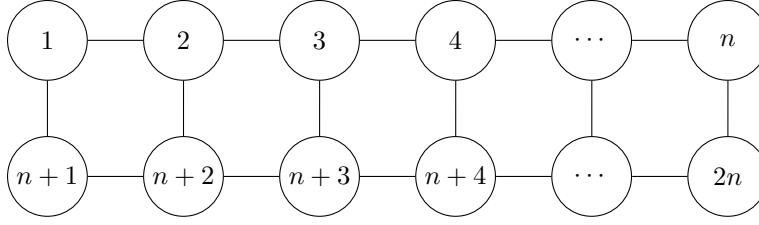
(b) G' , the resulting grid. G' is a $2 \times (n - (t - n) + 1)$ grid with corner vertices $V_c \in \{t - n, c, n, 2n\}$.

Figure 6

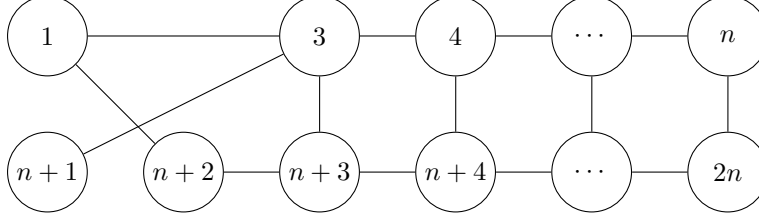
Proof. Let

$$G' = \tau_1 \circ X_{n+3} \circ Y_{n+2} \circ X_3 \circ X_2(G). \quad (13)$$

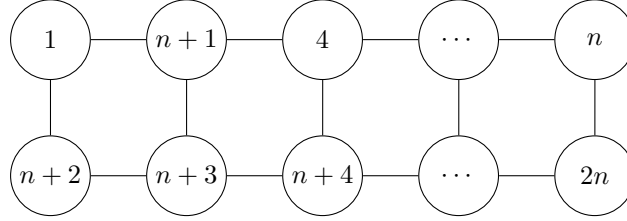
Use vertex 1 as the pivot for the X_2 measurement, vertex n as the pivot for the X_3 measurement, and vertex 1 as the pivot again for the X_{n+3} measurement. Then G' is a $2 \times (n - 2)$ grid where vertices 1 and $n + 1$ have swapped positions and vertices n and $2n$ have not changed. If n is even, $n - 2$ is also even, so one can extract $\{1, 2n\}$ and $\{n, n + 1\}$ from G' by Theorem 2.6 and therefore also from G . Similarly, if n is odd, one can extract $\{1, n\}$ and $\{n + 1, 2n\}$. $\square$



(a) G , a $2 \times n$ grid with $n > 4$



(b) $X_2(G)$



(c) $X_3 \circ X_2(G)$

Lemma 3.2. Let $G = (V, E)$ be a $2 \times n$ grid. Let $x \in V$ have $x \leq n$. Let $G' = (V', E')$. If $x = n$, let $V' = \{x, n+1\}$ and $E' = \{\{x, n+1\}\}$. Otherwise, let V' and E' satisfy the following:

$$V' = \{v \in V : x \leq v \leq n \text{ or } n+x+1 \leq v \leq 2n\} \cup \{n+1\} \quad (14)$$

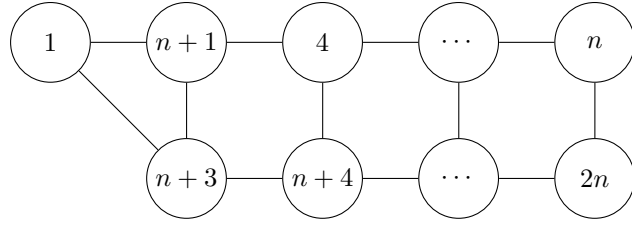
$$E' = \{\{v_1, v_2\} \in E : v_1, v_2 \in V'\} \cup \{\{x, n+1\}, \{n+1, n+x+1\}\}. \quad (15)$$

Then $G' < G$.

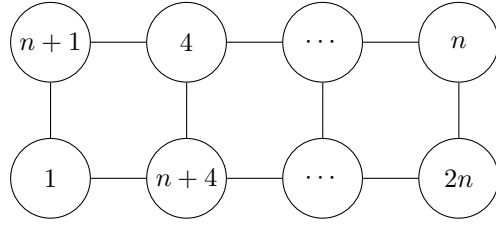
Proof. If $x = 1$, $G = G'$. Assume the lemma holds for $x = m$, where $m < n$. Then let $x = m+1$. Let

$$G^* = X_{n+2} \circ Z_1(G), \quad (16)$$

where vertex $n+1$ is used as the pivot for the X_{n+2} measurement. Then G^* is a $2 \times (n-1)$ grid with vertex $n+1$ unchanged, vertices 1 and $n+2$ deleted, and any other vertex v moved to the spot of $v-1$. By the inductive hypothesis, G^* can be transformed into G' .



(d) $Y_{n+2} \circ X_3 \circ X_2(G)$

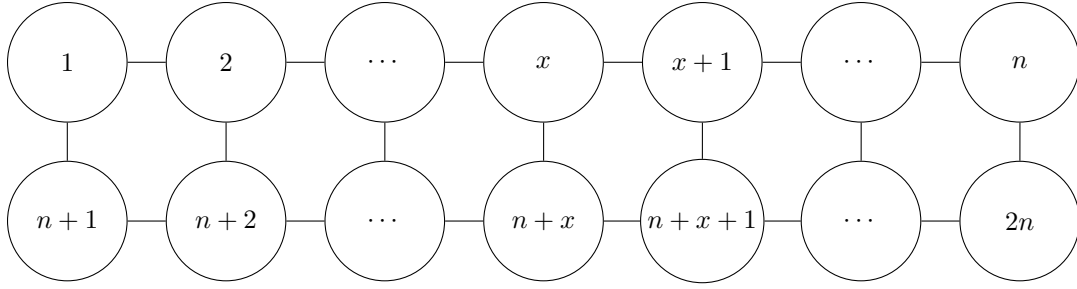


(e) $\tau_1 \circ X_{n+3} \circ Y_{n+2} \circ X_3 \circ X_2(G)$

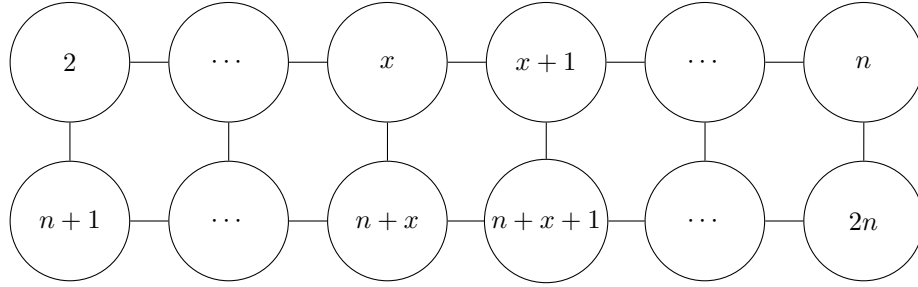
Figure 7

Therefore, $G' < G$.

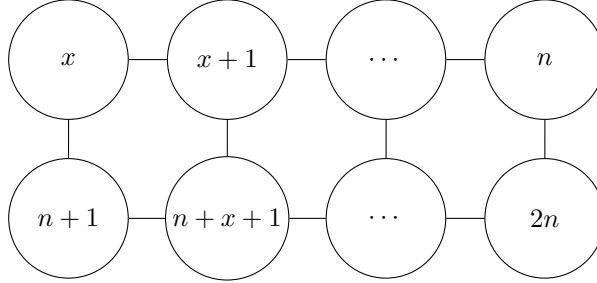
□



(a) G



(b) G^*



(c) G'

Figure 8

Theorem 3.1. Let $G = (V, E)$ be a $2 \times n$ grid. Suppose $\{a, b, c, d\} \subset V$ satisfy

$$n+1 \leq a < c \leq 2n \quad (17) \qquad a \leq b+n \quad (19)$$

$$1 \leq b < d \leq n \quad (18) \qquad c \leq d+n. \quad (20)$$

Let $x = c - b - n - 1$, the number of columns in between b and c . Using local Clifford operations, Pauli measurements, and classical communication, we have that:

- if $x = 1$, one can extract $\{a, d\}$ and $\{b, c\}$

- if $x = 2$, one can extract $\{a, c\}$ and $\{b, d\}$
- if $x \geq 3$, one can extract either $\{a, d\}$ and $\{b, c\}$ or $\{a, c\}$ and $\{b, d\}$.

Proof. Z-measure any vertices v satisfying $v \pmod n < a$ or $v \pmod n > d$. Then apply Lemma 3.2 twice, once with $x = b$ and once with $x = c$, this time using symmetry (temporarily relabel each vertex v to vertex $2n + 1 - v$) to reduce the other side of the grid. This yields a $2 \times (x + 2)$ grid.

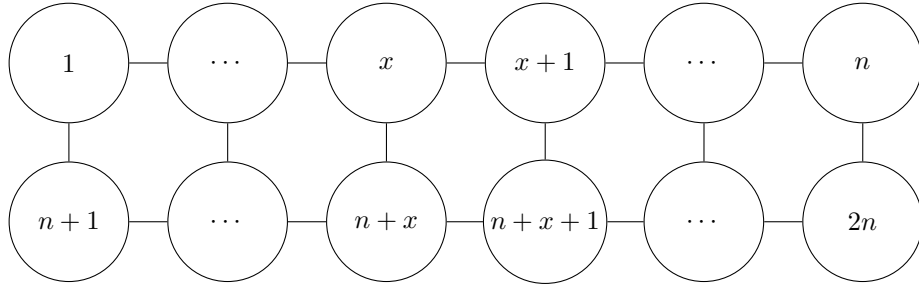
If $x = 1$, then extracting $\{a, d\}$ and $\{b, c\}$ is possible by Theorem 2.6. If $x = 2$, then extracting $\{a, c\}$ and $\{b, c\}$ is possible by Theorem 2.6. If $x \geq 3$, then both extracting $\{a, d\}$ and $\{b, c\}$ and extracting $\{a, c\}$ and $\{b, d\}$ are possible by Theorem 2.6 and Lemma 3.1. $\square$

Lemma 3.3. Let G be a $2 \times n$ grid, with $n \geq 4$ and $1 < x < n$. Then one can extract Bell pairs $\{x, n + x\}$ and $\{x + 1, n + x + 1\}$ using local Clifford operations, Pauli measurements, and classical communication. Alternatively, one can extract Bell pairs $\{x, x + 1\}$ and $\{n + x, n + x + 1\}$.

Proof. Z-measure any vertex satisfying $x \pmod n < x - 1$ or $x \pmod n > x + 2$. We are left with the 2×4 grid G' . Set

$$G^* = \tau_{x-1} \circ \tau_{n+x-1} \circ \tau_{x-1} \circ \tau_{x+2} \circ \tau_{n+x+2} \circ \tau_{x+2}(G') \quad (21)$$

$G^* < G$ is a ring on 6 vertices, where no 3 of the vertices $x, x + 1, n + x, n + x + 1$ are consecutive and there are non-intersecting paths from x to $n + x$ and from $x + 1$ to $n + x + 1$. By Theorem 2.5, one can extract $\{x, n + x\}$ and $\{x + 1, n + x + 1\}$. The paths from x to $x + 1$ and from $n + x$ to $n + x + 1$ do not intersect, so one can alternatively extract $\{x, x + 1\}$ and $\{n + x, n + x + 1\}$. $\square$



(a) G

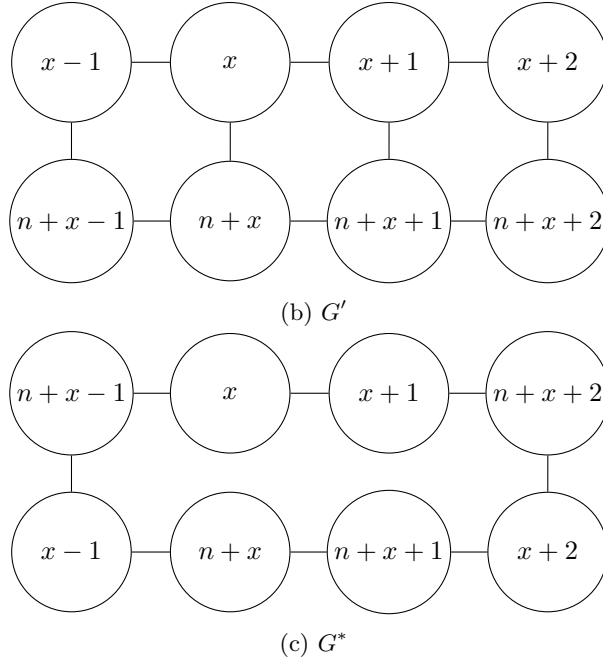


Figure 9

3.2 Local Sets and Rapport

Since the vertex minor relation preserves local sets by Corollary 2.1, by showing a set is a local set in one graph but not a local set in another, we can show the graphs are not vertex minor.

Lemma 3.4. There are no local sets of size 3 on $K_2 \cup K_2$.

Proof. Let S be a set of vertices of $K_2 \cup K_2$.

- If $|S| = 1$, then $|\text{Odd}(S)|$ must be 1, so the local set S generates is of size 2.
- If $|S| = 2$, then $|\text{Odd}(S)|$ must be 0 or 2, so the local set S generates is of size 2 or 4.
- If $|S| = 3$, then $|\text{Odd}(S)|$ must be 1, so the local set S generates is of size 4.

None of those have size 3. □

Corollary 3.1. Let $G = (V, E)$ be a graph and L a local set of size 3 on G . Then, for all $v \in V \setminus L$, there is no instance of $K_2 \cup K_2$ on the vertices $L \cup \{v\}$ that is vertex minor to G .

Now, we want to characterize the local sets of size 3.

Theorem 3.2. For $n \geq 3$, there are only 8 local sets of size 3 on a $2 \times n$ grid: the two highlighted in Figure 10 (S in blue, $\text{Odd}(S)$ in red), and their horizontal and vertical reflections.



Figure 10

Proof. Suppose $X = S \cup \text{Odd}(S)$ is a local set of size 3.

- If $|S| = 1$, then $|\text{Odd}(S)| = |N(S)| = 2$. The only vertices with 2 neighbors are the corners, in which case X is highlighted in Figure 10a.
- If $|S| = 2$, then $|\text{Odd}(S)| = 1$. If both vertices in S have 3 neighbors, since $|N(V_1) \cup N(V_2)| \leq 2$ for any two vertices on a grid, $\text{Odd}(S)$ would be at least 4, so one of the vertices must be in the corner. From there we can deduce it must li

□

As the proof of Corollary 3.1 shows, there just aren't that many local sets on $K_2 \cup K_2$. For that reason, we want to take a more fine-grained perspective on what is happening to these sets.

Definition 3.1. For any graph G , $S \subseteq V(G)$, and $v \in V(G)$, we say v has:

- **full rapport** with S if $v \in S \cap \text{Odd}_G(S)$
- **internal rapport** with S if $v \in (S - \text{Odd}_G(S))$
- **external rapport** with S if $v \in (\text{Odd}_G(S) - S)$
- **null rapport** or **no rapport** with S if $v \notin (S \cup \text{Odd}_G(S))$

The value of the rapport function $\text{raprt}_G(v, S)$ is the rapport between v and S in G .

Under this definition, every vertex has some kind of rapport with any S . Now, we see how it changes under local complementation

Lemma 3.5. For any graph G , $S \subseteq V(G)$, $v \in V(G)$, let

$$S_v = \begin{cases} S \setminus \{v\} & \text{if } v \in S \text{ and } v \in \text{Odd}_G(S) \\ S \cup \{v\} & \text{if } v \notin S \text{ and } v \in \text{Odd}_G(S) \\ S & \text{otherwise.} \end{cases}$$

Then, $S \cup \text{Odd}_G(S) = S_v \cup \text{Odd}_{\tau_v(G)}(S_v)$.

Proof. Proof of Theorem 2.7 in [6]. □

Theorem 3.3. For any graph G and vertices $u, v \in V(G)$, there is a single rapport permutation σ_v such that for all $S \subseteq V(G)$, $\text{raprt}_{\tau_v(G)}(u, S_v) = \sigma_v(\text{raprt}_G(u, S))$. Moreover, using cycle notation,

$$\sigma_v = \begin{cases} (\text{full external}) & \text{if } u = v \\ (\text{full internal}) & \text{if } u \in N(v) \\ () & \text{otherwise.} \end{cases}$$

Proof. Let $U = N_G(u)$ and $V = N_G(v)$.

Suppose $u \neq v$. Since we know $N_{\tau_v(G)}(u) = U \setminus V \cup V \setminus U$, through some set identities we can derive that for any $S \subseteq V(G)$

$$N_{\tau_v(G)}(u) \cap S = ((U \cap S_v) \setminus V) \cup ((V \cap S_v) \setminus U)$$

and

$$|N_{\tau_v(G)}(u) \cap S_v| = |U \cap S_v| + |V \cap S_v| - 2|U \cap V \cap S_v|. \quad (22)$$

If Equation 22 has the same parity as $|U \cap S|$, then $u \in \text{Odd}_{\tau_v(G)}(S)$ iff $u \in \text{Odd}_G(S)$; if it has different parity, $u \in \text{Odd}_{\tau_v(G)}(S_v)$ iff $u \notin \text{Odd}_G(S_v)$. Since the final term is always even, all we have to consider is whether $u \in \text{Odd}_G(S_v)$ and $v \in \text{Odd}_G(S_v)$.

First, if v has no rapport with S then $S_v = S$ and $|V \cap S|$ must be even, so $|U \cap S|$'s parity is preserved. If v has

□

Corollary 3.2. If $H < G$ then for every $S \subseteq V(H)$ there is a $S' \subseteq V(G)$ such that:

- for every $v \in V(H)$ there is a rapport permutation σ_v fixing null rapport which doesn't depend on S such that $\text{raprt}_H(v, S) = \sigma_v(\text{raprt}_G(v, S'))$.
- for every $u \in V(G) \setminus V(H)$ there is a non-null rapport r which doesn't depend on S such that $\text{raprt}_G(u, S')$ is either r or no rapport.

Proof. If $H < G$, that means that there is some $G' = \tau_{v_i} \circ \dots \circ \tau_{v_1}(G)$ such that $H = G' \setminus (V(G) \setminus V(H))$. Since local complementation is an involution, $G = \tau_{v_1} \circ \dots \circ \tau_{v_i}(G')$, so by repeatedly applying Theorem 3.3, for every $v \in V(G)$ there is a rapport permutation σ_v and for every $S \subseteq V(H) \subseteq V(G')$ there is a $S' \subseteq V(G)$ such that $\text{raprt}_{G'}(v, S') = \sigma_v(\text{raprt}_G(v, S))$. Since S is a subset of $V(H)$, if $v \in V(G') \setminus V(H)$ then it must have either external or no rapport with S in G' . Since σ_v fixes null rapport, v 's rapport with S' in G will be either null or $\sigma_v(\text{external})$. □

This corollary works nicely when determining if $K_2 \cup K_2 < G$: if we take a couple of subsets of one of the K_2 's we get two vertices in G that must have no rapport with the

lifted, two vertices that must have a variety of rapports, and the rest of the graph must have a specific rapport or no rapport.

Lemma 3.6. Let G be a $2 \times n$ grid. It is not possible to extract $\{1, 5\}$ and $\{2, x\}$, where $x \in [6, n]$, nor is it possible to extract $\{1, x\}$ and $\{2, 6\}$, where $x \in \{5\} \cup [7, n]$.

Proof. Suppose $\{1, 5\}$ and $\{2, x\}$ can be extracted from G . Let $D = \mathcal{P}(\{2, x\})$ be a family of generators on the $\{2, x\}$ component of $K_2 \cup K_2$ and D' be the result of lifting each element of D to G with Corollary 3.2. Note that 2 and x both achieve all 4 rapports among the sets in D , while 1 and 5 have no rapport with all of them. Therefore, 2 and x achieve all 4 rapports from D' , while 1 and 5 have no rapport with everything in D' . From the latter observation we can state the following: for any $S \in D'$:

$$2 \in S \iff 3 \in S$$

$$2 \in \text{Odd}_G(S) \iff 4 \in S \quad 3 \in \text{Odd}_G(S) \iff 4 \in S$$

By transitivity, we get $2 \in \text{Odd}_G(S) \iff 3 \in \text{Odd}_G(S)$. Together, this tells us that 2 and 3 have the same rapport with every element of D' . However, 2 gets all 4 rapports from D' , while 3 can only get 2. Therefore $\{1, 5\}$ and $\{2, x\}$ can't be extracted. By vertical symmetry of G , $\{2, 6\}$ and $\{1, x\}$ can't be extracted either. $\square$

This is in fact the final no-go lemma required to completely characterize which Bell pairs that involve the first column of a grid can be extracted.

Theorem 3.4. Let G be a $2 \times n$ grid, and $e, f \in V(G) \setminus \{1, 2\}$ be distinct vertices. Using local Clifford operations, Pauli measurements, and classical communication:

- one can extract $\{1, 2\}$ and $\{e, f\}$ iff $\{e, f\} \cap \{3, 4\} = \emptyset$.
- one can extract $\{1, e\}$ and $\{2, f\}$ iff $e \notin \{3, 4, 5\}$ and $f \notin \{3, 4, 6\}$.

Proof. For the first bullet point, by Theorem 3.2 if either e or f are in the second column then $\{1, 2\}$ and $\{e, f\}$ cannot be extracted, and if they aren't performing $Z_4 \circ Z_3$ splits the graph in two, at which point one just needs to extract $\{e, f\}$ by itself.

For the second bullet point, if $e \in \{3, 4\}$ or $f \in \{3, 4\}$ then by Theorem 3.2 it's not possible to extract $\{1, e\}$ and $\{2, f\}$. If $e = 5$ or $f = 6$ then by Lemma 3.6 it's not possible to extract $\{1, e\}$ and $\{2, f\}$. If $e = 6$ and $f \geq 5$ and odd, or if $f = 5$ and $e \geq 6$ and even, then it's possible to extract $\{1, e\}$ and $\{2, f\}$ by Theorem 3.1. If $e = 7$ and $f \geq 8$ and is even, or if $f = 8$ and $e \geq 7$ and is odd, then it's possible to extract $\{1, e\}$ and $\{2, f\}$ by Theorem 3.1. $\square$

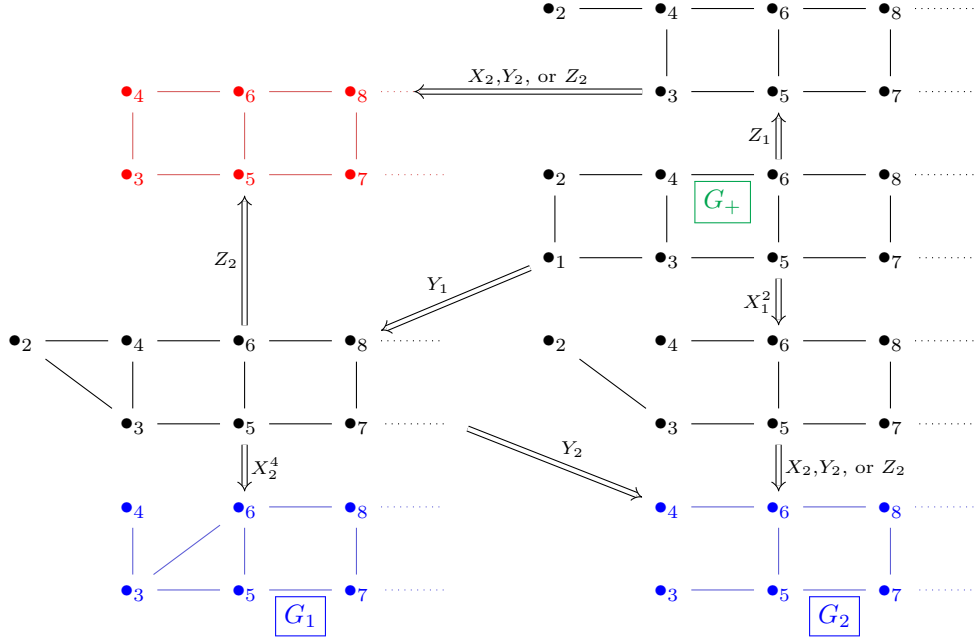


Figure 11

3.3 Adding a column

For the following section, we will want to visualize multiple graphs and the relationships between them at once. For this we use diagrams like Figure 11: vertices are circles labeled with numbers, and edges are single-width lines, and the dotted lines represent the graph extending arbitrarily long: it may not necessarily be a grid after that point. Each component is considered to be a different graph and the double arrows represent operations transforming one into the other, each of which is labeled by the transformation being applied.

Definition 3.2. A graph G has a $2 \times n$ grid as a **protrusion** iff it has a subgraph isomorphic to a $2 \times n$ grid and, for all vertices except $2n - 1$ and $2n$, there are no edges not generated by the grid.

Throughout this section, let the graph in Figure 11 labeled G_+ be a graph with a 2×3 grid as a protrusion and $a, b, c, d \notin \{1, 2\}$ be distinct vertices such that the graph H with vertices $\{a, b, c, d\}$ and edges $\{\{a, b\}, \{c, d\}\}$ is not vertex minor to $G_+ \setminus \{1, 2\}$.

Lemma 3.7. H is a vertex minor to G iff it is vertex minor to one of the graphs in Figure 11, G_1 and G_2 .

Proof. Since the vertices 1 and 2 are in $G_+ \setminus H$, by Theorem 2.1 $H < G$ iff there are $P_1 \in \{X_1, Y_1, Z_1\}$ and $P_2 \in \{X_2, Y_2, Z_2\}$ such that $H < P_2(P_1(G_+))$. In Figure 11 we

simply go through every combination of P_1 and P_2 . Since any two choice of pivots for X -measurement result in locally-equivalent graphs, we simply pick the pivot that is most convenient. Some combinations of P_1 and P_2 lead to the graph in red, $G_+ \setminus \{1, 2\}$: however H cannot be vertex minor to that graph would contradict our assumptions, so we can rule out those combinations. From this we can conclude that $H < G_0$ iff $H < G_1$ or $H < G_2$ $\square$

Lemma 3.8. If $H < G_+$ but $H \not< G_+ \setminus \{1, 2\}$, $\{3, 4\} \subset \{a, b, c, d\}$.

Proof. Suppose $4 \notin V(H)$. Then since 4 is a leaf in G_1 , by Theorem 2.2, $\hat{G} < G_1 \setminus \{4\}$, However, $G_1 \setminus \{4\}$ is a vertex minor of $G_+ \setminus \{1, 2\}$ via performing Y_4 , so $\hat{G} < G_+ \setminus \{1, 2\}$ contradicting our assumptions.

This argument also works when replacing the vertex 4 with 3 or G_1 with G_2 , except that 3 is an axil in G_1 . $\square$

Corollary 3.3. If $\{3, 4\} \not\subset V(\hat{G})$, then $H < G_+ \setminus \{1, 2\}$ iff $H < G_+$.

Proof.

$\Leftarrow$ If $H \not< G_+$ then by Lemma 3.8, $\{3, 4\} \subset \{a, b, c, d\}$, but that contradicts our assumptions.

$\Rightarrow$ Since $G_+ \setminus \{1, 2\} = Z_1 \circ Z_2(G_+)$

$\square$

Corollary 3.4 (Tail Reduction). Let G be a graph with a $2 \times n$ graph as a protrusion, where $n \geq 3$. Let $\text{col}(v)$ be the column of the vertex v in the protrusion, starting from the left (i.e. $\text{col}(v) = \lceil \frac{v}{2} \rceil$). Let $a, b, c, d \in V(G)$ be distinct, $\hat{G}$ be the graph with those as vertices and $\{\{a, b\}, \{c, d\}\}$ as edges, and $c = \text{col}(\min\{a, b, c, d\}) - 2$ Then, $\hat{G} < G \setminus [1, 2c]$ iff $\hat{G} < G$.

Proof. We will induct on n . For the base case where $n = 3$, since there are four vertices, $c \leq 0$, so $G \setminus [1, 0] = G$, so this is clearly a tautology.

For the inductive step, if $c = 0$ then $\min\{2c, 2(n-3)\} = 0$, so like in the base case this is a tautology. But if $c \geq 1$, then that implies that there are no vertices of $\hat{G}$ in the first two columns of G . By Corollary 3.3, $\hat{G} < G \setminus \{1, 2\}$ iff $\hat{G} < G$. But, since $G \setminus \{1, 2\}$ has a $2 \times (n-1)$ grid as a protrusion, and $n > 3$, we can use the inductive hypothesis to show that $\hat{G} < G_+ \setminus \{1, 2\}$ iff $\hat{G} < G \setminus [1, 2c]$. So by transitivity, $\hat{G} < G$ iff $\hat{G} < G \setminus [1, 2c]$. $\square$

By using Lemma 3.8 we can narrow down the possible pairs of pairings represented by H to the just the following: $H_1 = \{\{3, 4\}, \{e, f\}\}$ and $H_2 = \{\{3, e\}, \{4, f\}\}$, where $e, f \in [5, 2n]$.

Lemma 3.9. For any $e, f \in [5, 2n]$:

- (a) $H_2 \not\prec G_1$
- (b) If $H < G_2$ and $5 \in \{e, f\}$, then $H = H_2$ and $e = 5$.
- (c) If $H < G_2$ and $6 \in \{e, f\}$, then $H = H_2$ and $f = 6$.
- (d) For H_1 , $\{e, f\} \cap \{5, 6\} \neq \emptyset$

Proof.

- (a) $(4, 3)$ are a leaf-axil pair in G_1 , but are not foliage-equivalent in H_2 , contradicting Lemma 2.2.
- (b) Since $(3, 5)$ are a leaf-axil pair in G_2 , by Lemma 2.2 they must be foliage-equivalent in H .
- (c) Since $(4, 6)$ are a leaf-axil pair in G_2 , by Lemma 2.2 they must be foliage-equivalent in H .
- (d) Suppose neither e nor f were in the column $\{5, 6\}$. Then, H_1 would be vertex minor to $G_+ \setminus \{1, 2\}$ by splitting the grid using $Z_5 \circ Z_6$ and doing an X-protocol or something.

□

At this point, we can abandon the generality given to the previous lemmas which allowed G_+ to have a "tail" which might not be a grid. The flexibility will be useful in the proofs below, but all the future theorems and lemmas in the section will assume that, for $n \geq 4$, G_+ is just a $2 \times n$ grid, and so $G_1 \setminus \{3, 4\}$ and $G_2 \setminus \{3, 4\}$ are both $2 \times (n - 2)$ grids.

Lemma 3.10. H is a vertex minor of G_1 iff $H = H_1$ for some e, f and $\{e, f\} \notin \{\{5, 8\}, \{6, 7\}\}$.

Proof. By Lemma 3.9(a), pairings of the form H_2 cannot be vertex minor to G_1 , so H must equal H_1 for some e, f . Then, by Lemma 3.9(d), at least one of e and f must be in the second column of G_1 . WLOG say that it is e . The beautiful thing about this restriction is that no matter which f we pick, by Theorem 3.2, H_1 will not be vertex minor to $G_+ \setminus \{1, 2\}$. From here on our only worry is which choices of e and f allow H_1 to be vertex minor to G_1 . For now, let's say $e = 5$. We can break the f situation into 3 cases:

- $f = 6$: the operation $Y_7 \circ Y_8 \circ \tau_5$, followed by Z -measurement of every other vertex (if there are any), shows that $H_1 < G_1$. This is pictured in Figure 12a.
- $f = 8$: Since we require n to be at least 4, if $n > 4$ we use Corollary 3.4 so that we need to show is that H_1 is not vertex minor to G_1 restricted to the vertices $[3, 8]$. From there, we can apply Theorem 2.1 and enumerate all possible reductions. As shown in Figure 13, none of the possible reductions lead to H_1 , so $H_1 \not\prec G_1$.

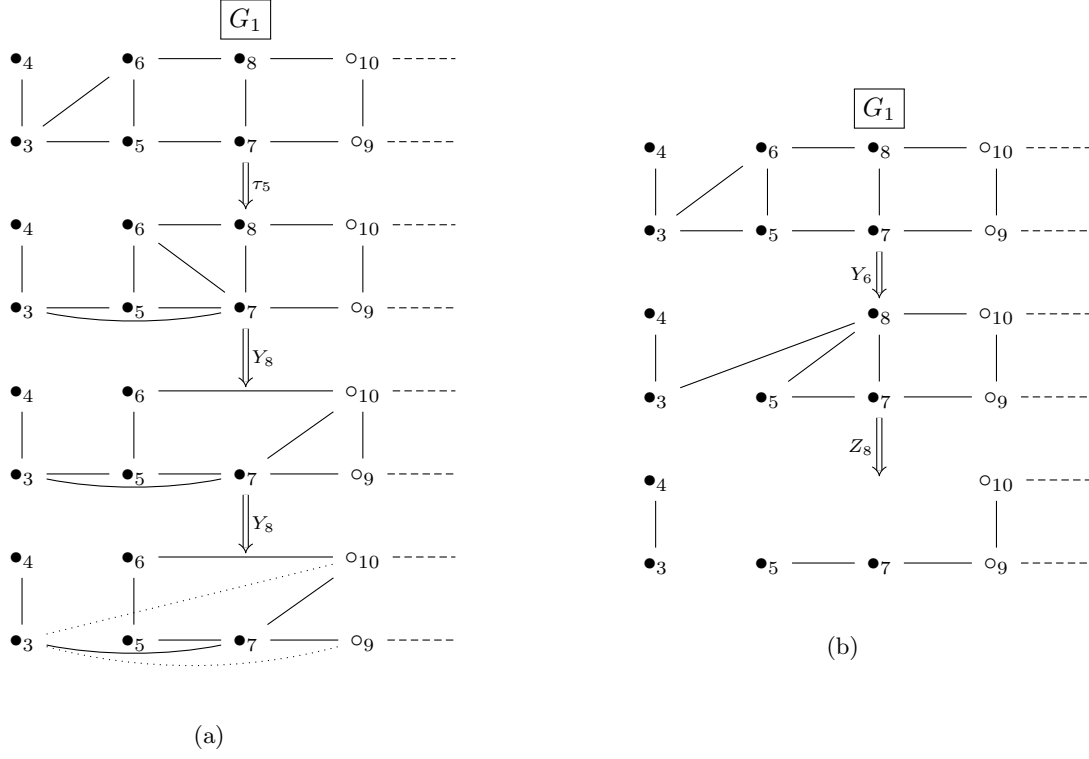


Figure 12

- $f \notin \{6, 8\}$: the operation $Z_8 \circ Y_6$ severs $\{3, 4\}$ from the rest of the graph. From there, we can connect e and f using an X-protocol or something, showing that $H_1 < G_1$. This is pictured in Figure 12b

We have just proven that if $e = 5$, then $H_1 < G_1$ iff $f \neq 8$. Due G_1 's vertical symmetry, if we let $e = 6$ the previous argument continues to work. The three cases become $f = 5$, $f = 7$, and $f = \text{"anything else"}$, and the conclusion is that $H_1 < G_1$ iff $f \neq 7$. Since these are the only two choices of e , we have completely characterized which H 's are vertex minor to G_1 $\square$

Moving on from the easy case, we lay out a couple of lemmas for working with G_2 . Lemma 3.12 provides an exact criterion for $H < G_2$ when e or f is in the second column, and Lemma 3.11 provides a reduction for when they aren't.

Lemma 3.11. Let T be the graph $G_+ \setminus \{1, 2, 5, 6\}$ but with the added edges $\{\{3, 7\}, \{4, 8\}\}$. If $\{5, 6\} \cap \{e, f\} = \emptyset$, then $H < G_2$ iff $H < T$

Proof. If neither 5 nor 6 are in H then we can apply Theorem 2.2 to them, telling us $H < G_2$ iff it is vertex minor to $\tau_3 \circ \tau_4(G_2) \setminus \{5, 6\}$, which is just T . $\square$

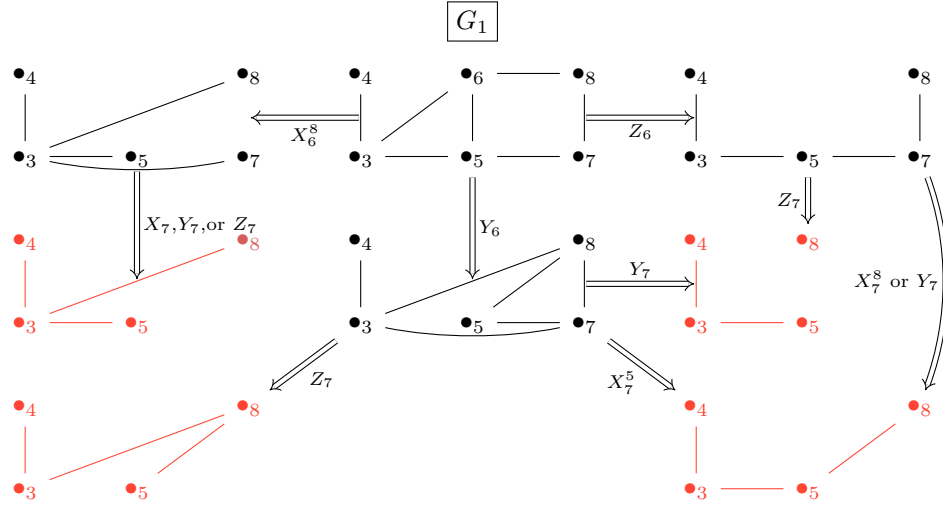


Figure 13

Lemma 3.12.

1. If $e = 5$, then $H < G_2$ iff $f \neq 7$.
2. If $f = 6$, then $H < G_2$ iff $e \neq 8$.

Lemma 3.13.

- (a) If $e = 5$, then $H < G_2$ iff $f \neq 8$.
- (b) If $f = 6$, then $H < G_2$ iff $e \neq 7$.

Proof.

- (a)
- (b) By vertical symmetry on G_2 .

□

Lemma 3.14. If $H < G_2$ and $\{e, f\} \cap \{5, 6\} = \emptyset$, then $\{e, f\} \cap \{7, 8\} = \emptyset$.

Proof. Since 5 and 6 are axils, we can apply Theorem 2.2 on both vertices. From there, $\{3, 4, 7\}$ and $\{3, 4, 8\}$ are local sets, so by Corollary 3.1, neither e nor f can be 7 or 8. □

3.4 Overcoming Line Graph Bottlenecks

Here we aim to resolve the line graph bottlenecks presented in [8] when the line graph is extended to grid. We were able to successfully resolve the "nested" bell pair case, and the "adjacent" bell pair case; However, more work will need to be done to determine if the "overlapping" case can be resolved. We suspect that if the line graph is extended to a $3 \times n$ grid, we could implement the crossing protocol to resolve some cases of "overlapping" bell pairs.

Lemma 3.15. Let $G = (V, E)$ be a $2 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$, where $kn + 1 \leq a_1 < b_1 < b_2 < a_2 \leq (k + 1)n$ and $b_2 = b_1 + 1$, for some $0 \leq k \leq 1$. See appendix C.1

Proof. Let $n \geq 4$, then by induction, we can show that bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

Base Case: $n = 4$. See figure Figure 28

If $n=4$, then $\{a_1, b_1, b_2, a_2\}$ must be next to each other, in that order. WLOG suppose $k=1$, so all the vertices are on the 2nd row.

Then $a_1 = 5$, $b_1 = 6$, $b_2 = 7$, and, $a_2 = 8$. Then $G' < G$ where G' contains the extracted bell pairs $\{5, 8\}$ and $\{6, 7\}$.

We can achieve G' by performing the following sequence of operations.

First perform local complementations on vertices $\{5, 8\}$. Then perform Y measurements on vertices $\{1, 4\}$. The resulting graph contain the extracted bell pair $\{6, 7\}$ and a line graph containing $\{5, 2, 3, 8\}$. By performing Y measurements on vertices $\{2, 3\}$, you can extract the second bell pair. $G' = Y_2 \circ Y_3 \circ Y_1 \circ Y_4 \circ \tau_8 \circ \tau_5(G)$

Inductive Step

Now suppose that Lemma 3.15 holds for $n \geq 4$. Consider the $2 \times (n + 1)$ grid and WLOG assume $k = 1$. Since $\{b_1, b_2\}$ must be neighbors and $n + 1 > 4$, there are 4 cases:

Case 1: See figure Figure 29a. Suppose $a_1 > (n + 1) + 1$. Then we know $\{a_2, b_1, b_2\}$ must all be greater than $(n + 1) + 1$ as well. We can safely perform Z measurements on the first column. Thus the resulting grid is a $2 \times n$ containing $\{a_1, a_2, b_1, b_2\}$, so by our inductive hypothesis $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

Case 2: See figure Figure 29c. Suppose $a_2 < 2(n + 1)$. Then we know $\{a_1, b_1, b_2\}$ must all be less than $2(n + 1)$ as well. We can safely perform Z measurements on the last column. Thus the resulting grid is a $2 \times n$ containing $\{a_1, a_2, b_1, b_2\}$, so by our inductive hypothesis $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

Case 3: See figure Figure 29e. Suppose $a_1 = (n + 1) + 1$, $a_2 = 2(n + 1)$ and $b_1 = a_1 + 1 = (n + 1) + 2$. Then we know $b_2 = (n + 1) + 3$ since b_1 and b_2 must be neighbors. We can see that b_2 belongs to the 3rd column. Since $a_2 = 2(n + 1)$, a_2 must belong to the $(n + 1)$ column. Then since $n \geq 4$, b_2 and a_2 cannot be neighbors.

We can then delete vertex $\{n+1\}$. Next perform a Y measurement on vertex $\{2(n+1)-1\}$. Finally by performing a local complementation on vertex $2(n+1)$, the resulting grid is $2 \times n$ grid containing $\{a_1, a_2, b_1, b_2\}$, so by our inductive hypothesis $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

Case 4: See figure Figure 29h. Suppose $a_1 = (n+1)+1$, $a_2 = 2(n+1)$ and $b_1 > a_1+1 = (n+1)+2$. Then we know that b_1 and a_1 cannot be neighbors. We also see that $\{a_2, b_2\}$ must be greater than b_1 , so they also do not neighbor a_1 .

We can then delete vertex $\{1\}$. Next perform a Y measurement on vertex $\{(n+1)+2\}$. Finally by performing a local complementation on vertex $\{a_1 = (n+1)+1\}$, the resulting grid is $2 \times n$ grid containing $\{a_1, a_2, b_1, b_2\}$, so by our inductive hypothesis $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

If $k=0$, you can use the same technique except performed on inverted vertices to show that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable. □

Lemma 3.16. Let $G = (V, E)$ be a $2 \times n$ grid. If $n \geq 6$, then it is possible to simultaneously extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$, where $kn+1 \leq a_1 < b_1 < b_2 < a_2 \leq (k+1)n$ and $b_2 > b_1+2$, for some $0 \leq k \leq 1$.

Proof. Let $n \geq 6$. WLOG assume that $\{a_1, a_2, b_1, b_2\}$ all lie on the bottom row, so $k=1$. Then G contains vertices such that $a_1 < b_1 < b_2 < a_2$ and $b_2 > b_1+2$. We can show that $G' < G$ where G' is a $2 \times m$ grid with corner vertices $V_c \in \{a_1, b_1, a_2, b_2\}$ and $m \geq 4$.

To do this we first perform vertex deletions so that the bottom corner vertices are $\{a_1, a_2\}$. This is achieved by performing Z measurements on all vertices in columns left of a_1 and right of a_2 .

Next, we will ensure that $\{a_1, b_1\}$ and $\{b_2, a_2\}$ are neighbors. If they are already neighbors then move onto the next step. If one, or both of them are not, use the corner replacement protocol on a_1 and/or a_2 : $\rho_{a_1}^{b_1-1}$ and/or $\rho_{a_2}^{b_2+1}$. See definition 2.13.

Now we have a grid such that $\{a_1, a_2\}$ are the bottom corner vertices. Additionally, $b_1 = a_1+1$ and $b_2 = a_2-1$. We can use the corner folding protocol, see def 2.14, to move $\{a_1, a_2\}$ to the top corners, while $\{b_1, b_2\}$ are turned into the bottom corners. We also note that the grid is a $2 \times m$ grid, where $m = b_2 - b_1 + 1$. Since $b_2 > b_1+2$, $m > 3$, so $m \geq 4$.

Thus G' is a $2 \times m$ grid with corner vertices $V_c \in \{a_1, b_1, a_2, b_2\}$ and $m \geq 4$. Then by Theorem 3.1 we can see that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

If $k=0$, you can use the same technique except performed on inverted vertices to show that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable. □

Lemma 3.17. Let $G = (V, E)$ be a $2 \times n$ grid. If $n \geq 5$, then it is possible to simultaneously extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$, where $kn+1 \leq a_1 < b_1 < b_2 < a_2 \leq (k+1)n$ and $b_2 = b_1+1$, for some $0 \leq k \leq 1$.

Proof. Let $n \geq 5$. WLOG assume that $\{a_1, a_2, b_1, b_2\}$ all lie on the bottom row, so $k = 1$. Then G contains vertices such that $a_1 < b_1 < b_2 < a_2$ and $b_2 = b_1 + 1$. We can show that $G' < G$ where G' contains 2 line graphs, each with a desired bell pair.

To do this we first perform vertex deletions so that the bottom corner vertices are $\{a_1, a_2\}$. This is achieved by performing Z measurements on all vertices in columns left of a_1 and right of a_2 .

Next, we will ensure that $\{a_1, b_1\}$ and $\{b_2, a_2\}$ are neighbors. If they are already neighbors then move onto the next step. If one, or both of them are not, use the corner replacement protocol on a_1 and/or a_2 : $\rho_{a_1}^{b_1-1}$ and/or $\rho_{a_2}^{b_2+1}$. See definition 2.13.

Now we have a grid such that $\{a_1, a_2\}$ are the bottom corner vertices. Additionally, $b_1 = a_1 + 1$ and $b_2 = a_2 - 1$. Since $b_2 = b_1 + 1$, we know that the graph is now a 2×5 grid.

Thus in the 2×5 grid, $a_1 = 6$, $a_2 = 10$, $b_1 = 7$, and $b_2 = 9$.

We can now use the following protocol on the grid to achieve G' . First perform a local complementation on vertex $\{8\}$. Then, perform X measurements on vertex $\{4\}$ using vertex $\{5\}$ as a pivot, and then on vertex $\{2\}$ using vertex $\{1\}$ as a pivot. Then perform a local complementation on vertex $\{5\}$ and vertex $\{1\}$. Finally, perform a Z measurement on vertex $\{1\}$, vertex $\{8\}$, and then vertex $\{5\}$. Then $G' = Z_5 \circ Z_8 \circ Z_1 \circ \tau_1 \circ \tau_5 \circ X_2^1 \circ X_4^5 \circ \tau_8(G)$. See appendix C.2.

Thus G' is a graph containing the extracted bell pair $\{b_1, b_2\}$ and the line graph $\{a_1, 3, a_2\}$. Then by performing a Y measurement on vertex $\{3\}$ we can see that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

If $k=0$, you can use the same technique except performed on inverted vertices to show that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable. $\square$

Theorem 3.5. Let $G = (V, E)$ be a $2 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$, where $kn + 1 \leq a_1 < b_1 < b_2 < a_2 \leq (k + 1)n$ for some $0 \leq k \leq 1$.

Proof. It is easy to see that this follows directly from Lemma 3.15, Lemma 3.16, and Lemma 3.17. $\square$

Lemma 3.18. Let $G = (V, E)$ be a $2 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$, where $kn + 1 \leq a_1 < a_2 < b_1 < b_2 \leq (k + 1)n$ and $b_1 = a_2 + 1$, for some $0 \leq k \leq 1$.

Proof. Let $n \geq 4$. WLOG assume that $\{a_1, a_2, b_1, b_2\}$ all lie on the bottom row, so $k = 1$. Then G contains vertices such that $a_1 < a_2 < b_1 < b_2$ and $b_1 = a_2 + 1$. We can show that $G' < G$ where G' is a 2×4 grid s.t $a_1 = 5$, $a_2 = 6$, $b_1 = 7$, and $b_2 = 8$.

To do this we first perform vertex deletions so that the bottom corner vertices are $\{a_1, b_2\}$. This is achieved by performing Z measurements on all vertices in columns left of a_1 and right of b_2 .

Next, we will ensure that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are neighbors. If they are already neighbors then move onto the next step. If one, or both of them are not, use the corner replacement protocol on a_1 and/or b_2 : $\rho_{a_1}^{a_2-1}$ and/or $\rho_{b_2}^{b_1+1}$. See definition 2.13.

Now we have a grid such that $\{a_1, b_2\}$ are the bottom corner vertices. Additionally, $a_1 = a_2 - 1$ and $b_2 = b_1 + 1$. Since $b_1 = a_2 + 1$, we know that the graph is now the 2×4 grid.

Thus in the 2×4 grid, $a_1 = 5$, $a_2 = 6$, $b_1 = 7$, and $b_2 = 8$.

We can now use the following protocol on G' to simultaneously extract $\{a_1, a_2\}$ and $\{b_1, b_2\}$. First perform Y measurements on vertex $\{2\}$, and then on vertex $\{3\}$. Finally, perform a Z measurement on vertex $\{1\}$ and then vertex $\{4\}$. Then $G'' = Z_4 \circ Z_1 \circ Y_3 \circ Y_2(G')$. See appendix C.3

Thus G'' contains the extracted bell pairs. So, we can see that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable.

If $k=0$, you can use the same technique except performed on inverted vertices to show that $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are simultaneously extractable. $\square$

It is not hard to see that since all these cases work for a $2 \times n$ grid, one could easily extend them to a $m \times n$ grid.

4 $3 \times n$ Grids

In this section we look at constructions for $3 \times n$ grids. We focus on the side border cases where the vertices of the bell pairs lie in either the first column, or the last column. We determined that for all possible arrangements of bell pairs, we can simultaneously extract them.

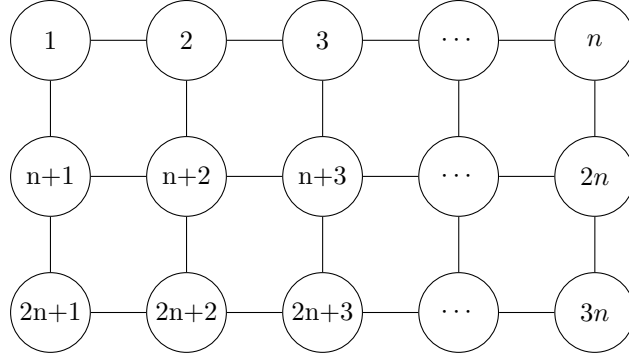
4.1 Non-Crossing Bell Pairs

Lemma 4.1. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 2$, then it is possible to simultaneously extract Bell pairs $\{1, n\}$ and $\{2n+1, 3n\}$. See Figure 14

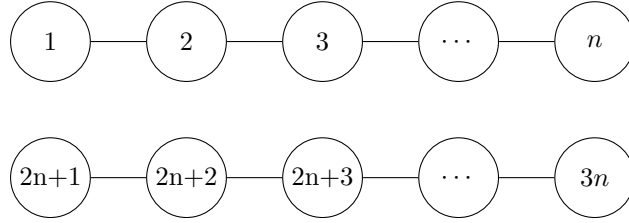
Proof. Let $n \geq 2$, then $G' < G$, where G' consists of two separate line graphs, each containing one bell pair.

First, perform Z measurements on the middle row so that $G' = Z_{2n} \circ \dots \circ Z_{n+1}(G)$. The resulting graph is two separate line graphs with one containing $\{1, n\}$ as end points and the other containing $\{2n+1, 3n\}$ as end points.

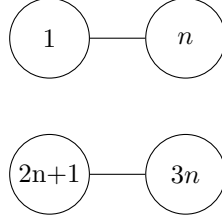
Then simply perform Y measurements on all intermediary vertices, so that $G'' = Y_{n-1} \circ \dots \circ Y_2 \circ Y_{3n-1} \circ \dots \circ Y_{2n+2}(G')$. $\square$



(a) G is a $3 \times n$ grid where $n \geq 2$. We will simultaneously extract $\{1, n\}$ and $\{2n+1, 3n\}$.



(b) Perform Z measurements along all vertices in the middle row.



(c) Perform Y measurements on all vertices except $\{1, n, 2n+1, 3n\}$.

Figure 14

Lemma 4.2. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{1, n\}$ and $\{n+1, 2n\}$.

Proof. Let $n \geq 4$, then it is easy to see that $G' < G$, where G' is a $2 \times n$ graph, with corner vertices $V_c \in \{1, n, n+1, 2n\}$. We can achieve this by performing Z measurements along the 3rd row.

By Theorem 3.1, $\{1, n\}$ and $\{n+1, 2n\}$ are simultaneously extractable.

□

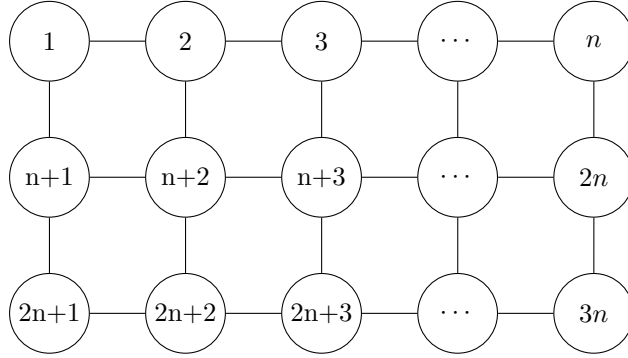
Lemma 4.3. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{1, n\}$ and $\{2n+1, 2n\}$. See Figure 15

Proof. Let $n \geq 4$, then $G' < G$, where G' is a $2 \times n$ graph, with corner vertices $V_c \in \{1, n, 2n+1, 2n\}$.

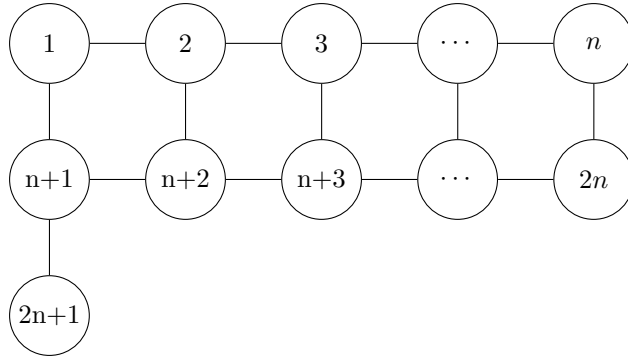
Begin by performing vertex deletions on all vertices on the bottom row, except vertex $\{2n+1\}$. Then, $G_1 = Z_{3n} \circ \dots \circ Z_{2n+2}(G)$ looks like a $2 \times n$ grid with the vertex $\{2n+1\}$ being a leaf attached to the corner vertex, $V_c = \{n+1\}$, of the $2 \times n$ grid.

Next, perform a Y measurement on vertex $\{n+1\}$. Notice, that the only neighbors of $\{n+1\}$ are vertices $\{1, n+2, 2n+1\}$. This results in $G_2 = Y_{n+1}(G_1)$ looking like a $2 \times n$ grid with corner vertices $V_c = \{1, n, 2n+1, 2n\}$ and an additional edge $(1, n+2)$.

Finally, perform a local complementation on $\{2n+1\}$ to get rid of that extra edge. Then $G' = \tau_{2n+1}(G_2)$, which is a $2 \times n$ grid with corner vertices $V_c = \{1, n, 2n+1, 2n\}$. By Theorem 3.1, Bell Pairs $\{1, n\}$ and $\{2n+1, 2n\}$ are simultaneously extractable for $n \geq 4$. $\square$



(a) G is a $3 \times n$ grid where $n \geq 4$. We will simultaneously extract $\{1, n\}$ and $\{2n+1, 2n\}$.



(b) $G_1 = Z_{3n} \circ \dots \circ Z_{2n+2}(G)$; Perform Z measurements along all vertices in the bottom row except $\{2n+1\}$.

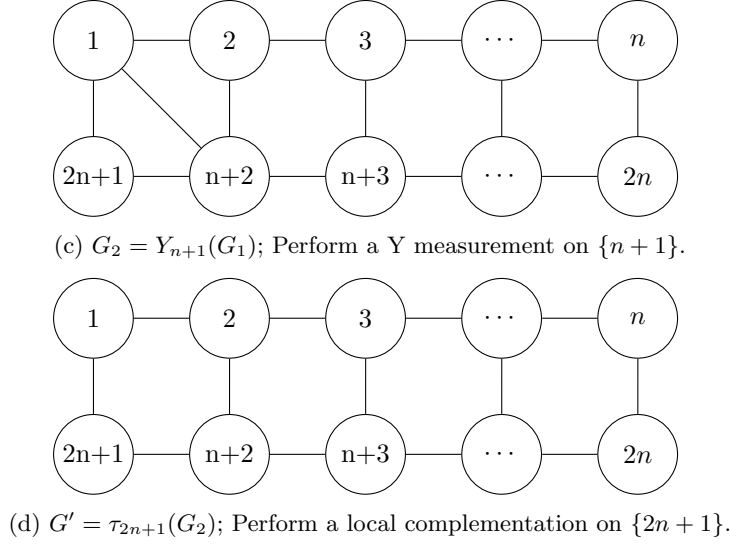


Figure 15

Lemma 4.4. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 4$, then it is possible to simultaneously extract Bell pairs $\{1, n\}$ and $\{n+1, 2n+1\}$. See Figure 16

Proof. Let $n \geq 4$, then $G' < G$, where G' consists of two separate line graphs, each containing one bell pair.

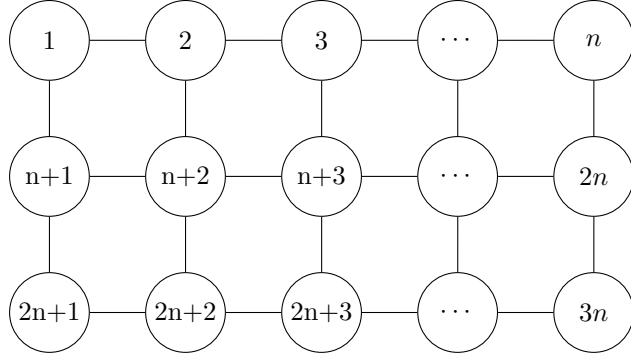
Begin by performing the crossing protocol on vertex $\{n+2\}$ so that $G_1 = \chi_{n+2}(G)$. The resulting graph has vertex $\{1\}$ connected to vertex $\{n+3\}$ as a leaf which is then connected to the greater grid. Additionally, vertex $\{n+1\}$ is connected to vertex $\{3\}$ as a leaf which is then connected to the greater grid. It is also important to note that vertex $\{2n+1\}$ is attached to $\{n+1\}$ as a leaf.

Next, perform a Z measurement on vertex $\{3\}$. Notice, that we have now extracted the $\{n+1, 2n+1\}$ bell pair. This results in $G_2 = Z_3(G_1)$ containing one extracted bell pair.

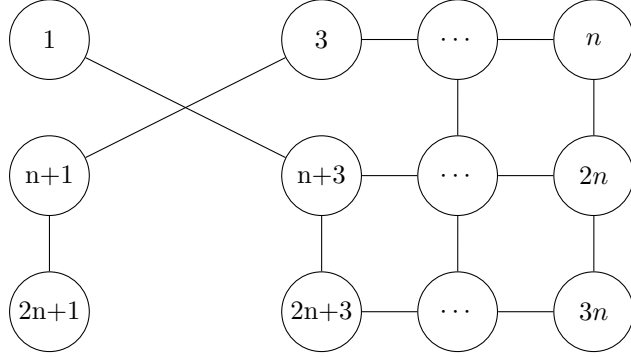
It is not hard to see that by choosing any path between vertices $\{1, n\}$ and performing Z measurements on the vertices not in that path, you can extract a line graph with $\{1, n\}$ as the end points, G' .

Finally, by performing Y measurement on all vertices on the path between $\{1, n\}$, you can extract the second bell pair. Thus Bell Pairs $\{1, n\}$ and $\{2n+1, 2n\}$ are simultaneously extractable for $n \geq 4$. $\square$

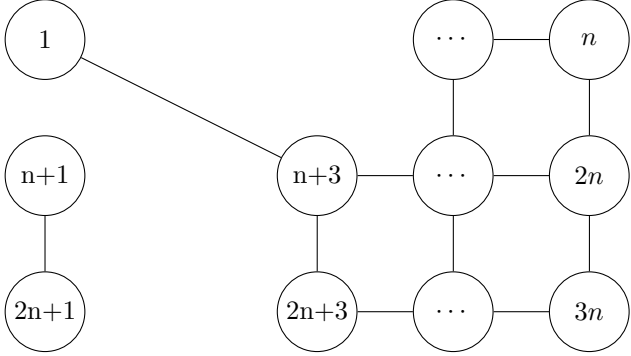
*Note: this solution also works for simultaneously extracting $\{1, 2n\}$ and $\{n+1, 2n+1\}$, and $\{1, 3n\}$ and $\{n+1, 2n+1\}$, for $n \geq 3$.



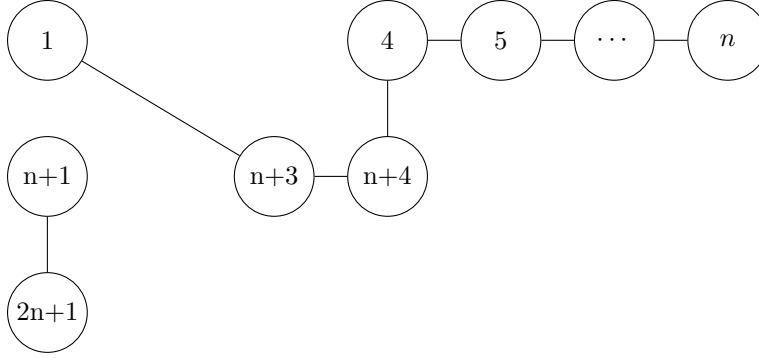
(a) G is a $3 \times n$ grid where $n \geq 4$. We will simultaneously extract $\{1, n\}$ and $\{n+1, 2n+1\}$.



(b) $G_1 = \chi_{n+2}(G)$; Perform the crossing protocol on $\{n+2\}$. See Figure 2



(c) $G_2 = Z_3(G_1)$; Perform a Z measurement on vertex $\{3\}$.



(d) G' ; Choose a path between $\{1, n\}$ and perform Z measurements on remaining nodes in the grid that are not on the path.

Figure 16

We believe that for certain cases these lemmas could be extended to an $m \times n$ grid; However, more work will need to be done to determine cases where these protocols could not be used in the extended grid.

4.2 Crossing Bell Pairs

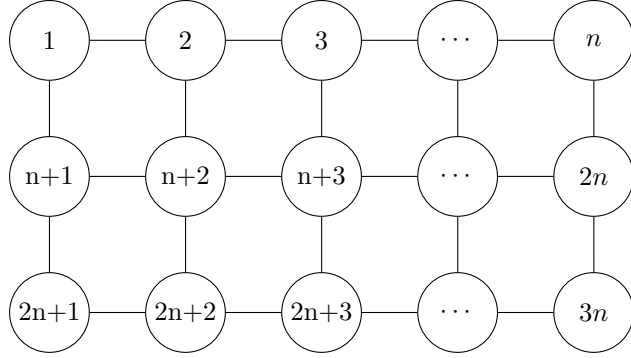
Lemma 4.5. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 3$, then it is possible to simultaneously extract Bell pairs $\{1, 3n\}$ and $\{n, 2n + 1\}$. See Figure 17

Proof. Let $n \geq 3$, then $G' < G$, where G' consists of two separate line graphs, each containing one bell pair.

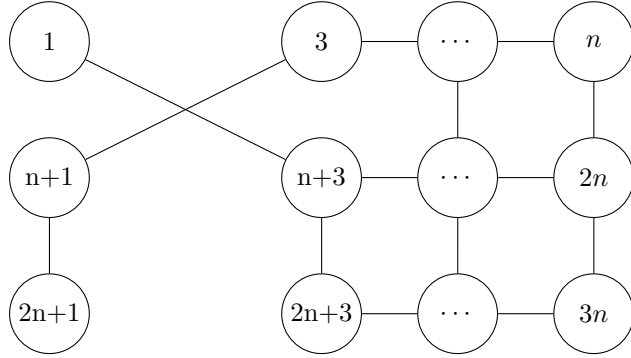
Begin by performing the crossing protocol on vertex $\{n + 2\}$ so that $G_1 = \chi_{n+2}(G)$. The resulting graph has vertex $\{1\}$ connected to vertex $\{n + 3\}$ as a leaf which is then connected to the greater grid. Additionally, vertex $\{n + 1\}$ is connected to vertex $\{3\}$ as a leaf which is then connected to the greater grid. It is also important to note that vertex $\{2n + 1\}$ is attached to $\{n + 1\}$ as a leaf. Finally, as a result of the crossing protocol, there is no edge between $(3, n + 3)$

It is not hard to see that by performing Z measurements on all the vertices in the middle row, right of vertex $\{n + 3\}$, you can extract two line graphs with $\{1, 3n\}$ and $\{n, 2n + 1\}$ as the end points, $G' = Z_{2n} \circ \dots \circ Z_{n+4}(G_1)$.

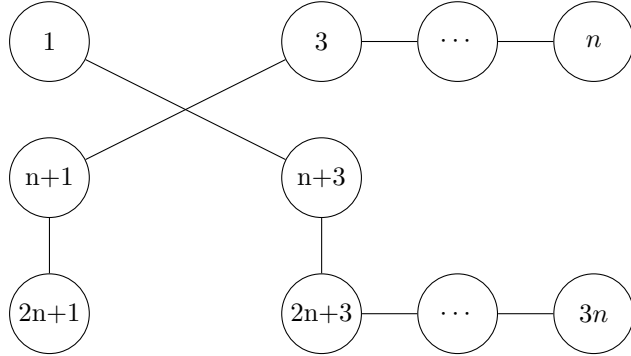
Finally, by performing Y measurement on all vertices on the path between $\{1, 3n\}$ and $\{n, 2n + 1\}$, you can simultaneously extract $\{1, 3n\}$ and $\{n, 2n + 1\}$. $\square$



(a) G is a $3 \times n$ grid where $n \geq 3$. We will simultaneously extract $\{1, 3n\}$ and $\{n, 2n+1\}$.



(b) $G_1 = \chi_{n+2}(G)$; Perform the crossing protocol on $\{n+2\}$. See Figure 2.



(c) $G' = Z_{2n} \circ \dots \circ Z_{n+4}(G_1)$; Perform Z measurements on middle row vertices, to the right of vertex $\{n+3\}$.

Figure 17

Lemma 4.6. Let $G = (V, E)$ be a $3 \times n$ grid. If $n = 3$ or $n \geq 5$, then it is possible to simultaneously extract Bell pairs $\{1, 2n\}$ and $\{n+1, n\}$.

Proof. Let $n = 3$ or $n \geq 5$, then it is easy to see that $G' < G$, where G' is a $2 \times n$ graph, with

corner vertices $V_c \in \{1, n, n+1, 2n\}$. We can achieve this by performing Z measurements along the 3rd row.

Using Theorem 3.1, $\{1, 2n\}$ and $\{n+1, n\}$ are simultaneously extractable. $\square$

Lemma 4.7. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 3$, then it is possible to simultaneously extract Bell pairs $\{1, 2n\}$ and $\{2n+1, n\}$.

Proof. Let $n \geq 3$.

If the graph is a 3×4 grid, before proceeding turn the graph into a 3×3 grid that preserves bell pair vertices. This can be done by deleting vertex $\{n+1\}$. Then perform Y measurements on vertices $\{2, 2n+2\}$. Finally perform local complementations on $\{1, 2n+1\}$. So $G_1 = \tau_{2n+1} \circ \tau_1 \circ Y_{2n+2} \circ Y_1 \circ Z_{n+1}(G)$ and G_1 is a 3×3 grid that preserves all the positions of the bell pair vertices.

Now all the $3 \times n$ grids have $n = 3$ or $n \geq 5$ columns, then $G' < G$, where G' is a $2 \times n$ graph, with corner vertices $V_c \in \{1, n, 2n+1, 2n\}$.

By performing the same sequence of local complementations and vertex deletions as in Lemma 4.3, we can achieve the graph G' .

Then G' is a $2 \times n$ grid with corner vertices $V_c = \{1, n, 2n+1, 2n\}$. By Theorem 3.1, Bell Pairs $\{1, 2n\}$ and $\{2n+1, n\}$ are simultaneously extractable for $n \geq 3$. $\square$

*Note: using a similar technique described here, we can see that for $n \geq 3$ you can also simultaneously extract: $\{1, 3n\}$ and $\{2n+1, 2n\}$; $\{1, 3n\}$ and $\{n+1, n\}$; $\{n+1, 3n\}$ and $\{2n+1, n\}$

Lemma 4.8. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 6$, then it is possible to simultaneously extract Bell pairs $\{1, 3n\}$ and $\{n+1, 2n\}$. See Figure 18

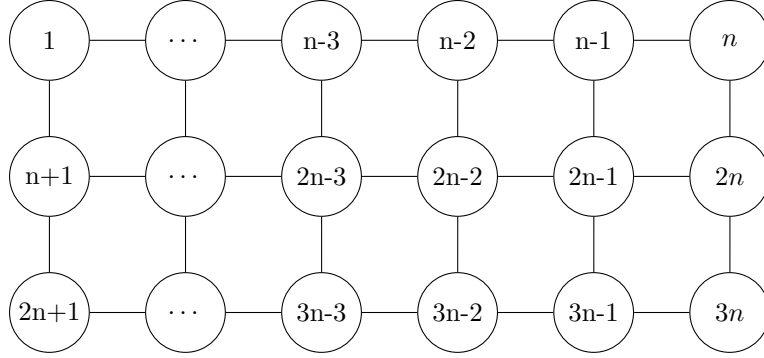
Proof. Let $n \geq 6$, then $G' < G$, where G' is a $3 \times (n-2)$ graph. This graph has the corner vertices $\{n, 3n\}$ flipped and columns $n-1$ and $n-2$ deleted. Finally the column n is shifted over and joined to the rest of the grid.

By performing the flipping protocol as defined in def2.12, we can achieve the graph G' .

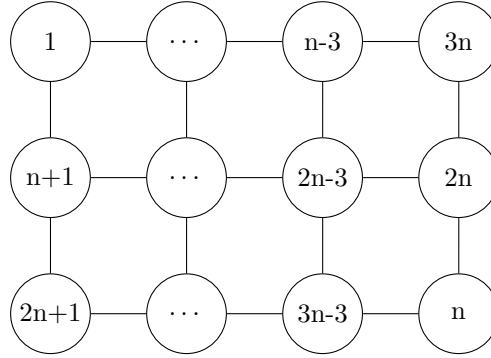
Thus by Lemma 4.2, since G' is $3 \times n-2$ grid where $n-2 \geq 4$, the bell pairs $\{1, 3n\}$ and $\{n+1, 2n\}$ are simultaneously extractable. $\square$

*Note: using a similar technique, if $n \geq 6$, then it is possible to simultaneously extract $\{n, 2n+1\}$ and $\{n+1, 2n\}$.

**Note: It is still not certain whether simultaneous extraction of $\{1, 3n\}$ and $\{n+1, 2n\}$ is possible for the $n = 3$, $n = 4$, and $n = 5$ cases.



(a) G is a $3 \times n$ grid where $n \geq 6$. We will simultaneously extract $\{1, 3n\}$ and $\{n+1, 2n\}$.



(b) $G' = \varphi(G)$; Perform the flipping protocol. Then by Lemma 4.2 bell pairs $\{1, 3n\}$ and $\{n+1, 2n\}$ are simultaneously extractable.

Figure 18

Lemma 4.9. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 3$, then it is possible to simultaneously extract Bell pairs $\{1, 2n+1\}$ and $\{n+1, n\}$.

Proof. Let $n \geq 3$, then $G' < G$.

Case 1: $n = 3$

If $n = 3$, then perform the corner extraction protocol, see def 2.12. Then G' contains the two extracted bell pairs $\{1, 7\}$ and $\{4, 3\}$. See Figure 4b

Case 2: $n = 4$

If $n = 4$, then perform the corner extraction protocol, see def 2.12. Then G' contains the extracted bell pair $\{1, 9\}$. Additionally, it contains a line graph with $\{5, 4\}$ as the endpoints. By performing a Y measurement on vertex $\{3\}$, the second bell pair can be extracted. See Figure 4c.

Case 3: $n > 4$

If $n > 4$, then perform the corner extraction protocol, see def 2.12. Then G' contains

the extracted bell pair $\{1, 2n + 1\}$. Additionally, it contains the line graph with vertices $\{n + 1, 3, 4\}$ in that respective order. The line graph is connected to a $2 \times n - 4$ grid via the edge $(4, 5)$. Finally, the $2 \times n - 4$ grid also contains the vertex n . See Figure 4c.

Select a path between $\{n + 1, n\}$ and then delete all remaining vertices in the grid. Then perform Y measurements along the path to extract the bell pair $\{n + 1, n\}$.

Thus it is possible to simultaneously extract Bell pairs $\{1, 2n + 1\}$ and $\{n + 1, n\}$

□

*Note: using a similar technique, if $n \geq 3$, then it is possible to simultaneously extract $\{1, 2n + 1\}$ and $\{n + 1, 3n\}$

Lemma 4.10. Let $G = (V, E)$ be a $3 \times n$ grid. If $n \geq 5$, then it is possible to simultaneously extract Bell pairs $\{1, 2n + 1\}$ and $\{n + 1, 2n\}$.

Proof. Let $n \geq 5$, then $G' < G$, where G' contains the extracted bell pair $\{1, 2n + 1\}$. Additionally, it contains the line graph with vertices $\{n + 1, 3, 4\}$ in that respective order. The line graph is connected to a $2 \times n - 4$ grid, which contains vertex $\{2n\}$, via the edge $(4, 5)$. See Figure 4c.

By performing the corner extraction protocol, see def 2.12, we can achieve G' .

Select a path between $\{n + 1, 2n\}$ and then delete all remaining vertices in the grid. Then perform Y measurements along the path to extract the bell pair $\{n + 1, 2n\}$.

Thus it is possible to simultaneously extract Bell pairs $\{1, 2n + 1\}$ and $\{n + 1, 2n\}$

□

5 Tadpole Graphs and Conjoined Rings

5.1 Tadpole Graphs

Definition 5.1. Let the tadpole graph $T_{m,n}(V_{m,n}, E_{m,n})$ have $V_{m,n} = \{1, 2, \dots, m + n\}$ and $E_{m,n} = \{\{1, 2\}, \{2, 3\}, \dots, \{m, 1\}\} \cup \{\{m, m + 1\}, \{m + 1, m + 2\}, \dots, \{m + n - 1, m + n\}\}$.

We solve the tadpole graph by looking at each choice of Bell pairs such that all, 3, 2, 1, or none of the vertices in the Bell pairs are in the ring portion of the tadpole graph.

Lemma 5.1. Let $a_1, a_2, b_1, b_2 \leq m$ be vertices in $T_{m,n}$. One can extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ using local Clifford operations, Pauli measurements, and classical communication iff no 3 of $\{a_1, a_2, b_1, b_2\}$ are consecutive and $a_1 < a_2 < b_1 < b_2$ (up to some reordering).

Proof. By repeated application of source reduction to delete every vertex $v > m$, the problem is reduced to extracting the Bell pairs from R_m . The result follows from Theorem 2.5. □

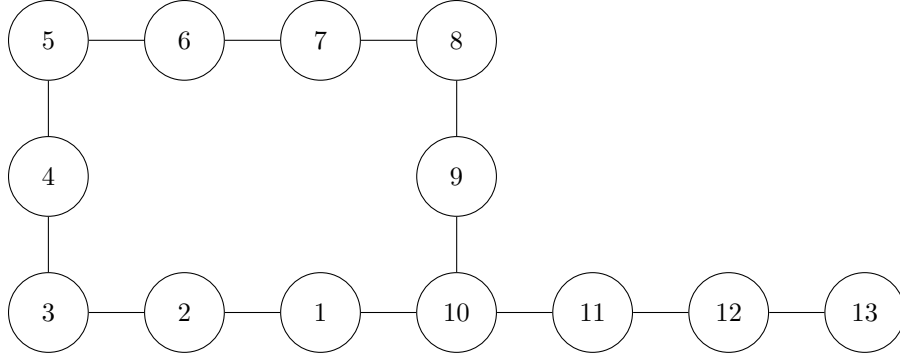


Figure 19: $T_{10,3}$

Lemma 5.2. Let $a_1 > m$ and $a_2, b_1, b_2 \leq m$ be vertices in $T_{m,n}$. One extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ using local Clifford operations, Pauli measurements, and classical communication iff either (1) $a_2 = m$ and a_2, b_1, b_2 are not consecutive in that order or (2) $m \neq a_2, b_1, b_2$, no 3 of $\{m, a_2, b_1, b_2\}$ are consecutive, and $m < a_2 < b_1 < b_2$ (up to some reordering).

Proof. Apply source reduction every vertex $v \neq a_1$ with $v > m$ until a_1 is the leaf of m .

Suppose b_1 or $b_2 = m$. The Bell pairs cannot be extracted; otherwise, this would contradict Lemma 2.2, because $a_1 \sim_F b_1$ or $a_1 \sim_F b_2$, but none of the descriptions correspond to the vertices in $K_2 \cup K_2$.

Suppose $a_2 = m$. If b_1, a_2, b_2 are consecutive in that order, Y-measure all but one vertex between b_1 and b_2 and X-measure the remaining vertex to obtain the isolated Bell pairs. If a_2, b_1, b_2 are not consecutive in some order, one can extract the target graph by Y measuring any vertices between b_1 and b_2 and deleting any other vertices besides a_2, b_1 , and b_2 . Suppose a_2, b_1, b_2 are consecutive, in that order. By target reduction, it is possible to extract the Bell pairs only if one can extract the pair $\{b_1, b_2\}$ plus the isolated vertex a_2 from R_m . This is impossible by Lemma 2.3.

Suppose $m \neq a_2, b_1, b_2$. Apply source reduction to the axil of a_1 to reduce the problem to extracting the Bell pairs from R_m , where a_1 has taken the place of vertex m . By Theorem 2.5, one can extract the Bell pairs in iff $m < a_2 < b_1 < b_2$ (up to some reordering) and no 3 of $\{m, a_2, b_1, b_2\}$ are consecutive in $T_{m,n}$. $\square$

Lemma 5.3. Let a_1, a_2, b_1, b_2 be vertices of $T_{m,n}$ such that exactly 2 vertices $v_1, v_2 \in \{a_1, a_2, b_1, b_2\}$ satisfy $v_1, v_2 \leq m$. One can extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ using local Clifford operations, Pauli measurements, and classical communication iff $v_1 = a_1$, $v_2 = a_2$, and a_1 and a_2 are not adjacent to b_1 or b_2 .

Proof. WLOG, assume $a_1 > m$. Suppose $b_1 > m$. Apply source reduction repeatedly to delete all vertices $v > m$ except a_1 and b_1 . a_1 becomes either the leaf or axil of b_1 .

If $K_2 \cup K_2 < T_{m,n}$, this would contradict Lemma 2.2, since $a_1 \sim_F b_1$ but none of the descriptions correspond to $\{a_1, b_1\}$ in $K_2 \cup K_2$.

Otherwise, $a_1, a_2 > m$. Apply source reduction until, WLOG, a_1 is the leaf of a_2 . Call the new graph T' . If a_2 is not adjacent to b_1 or b_2 , then delete any other vertex $v > m$ except for a_1 and, if $b_1, b_2 \neq x$, delete vertex m . Then, Y-measure any vertices between b_1 and b_2 and delete any other vertex $u \leq m$ to obtain $K_2 \cup K_2$.

Suppose a_2 is adjacent to b_1 . By target reduction, it is possible to extract the Bell pairs only if it is possible to extract the pair $\{b_1, b_2\}$ plus the isolated vertex a_2 from the graph $Z_{a_1}(T')$. Apply source reduction to delete any other vertex $w > m$ until a_2 becomes the leaf or axil of b_1 . Extraction is impossible; if it were possible, this would contradict Lemma 2.2, since $a_1 \sim_F b_1$ but none of the descriptions correspond to $\{a_1, b_1\}$ in $K_2 \cup K_2$. $\square$

Lemma 5.4. Let $a_1, a_2, b_1 > m$ and $b_2 \leq m$ be vertices in $T_{m,n}$. Then one can extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ using local Clifford operations, Pauli measurements, and classical communication iff $a_1 > a_2 > b_1$ (up to some reordering), and a_2 is not adjacent to b_1 .

Proof. Apply source reduction to delete any vertices $v > a_1, a_2, b_1$ and call the new graph T' . Suppose $a_1 > a_2 > b_1$ and a_2 and b_1 are not adjacent. One can extract the Bell pairs by deleting all vertices in between a_2 and b_1 , Y-measuring any vertices v satisfying $b_2 < v < b_1$, and then deleting any remaining vertices not in the Bell pairs.

Suppose $a_1 > a_2 > b_1$ but a_2 and b_1 are adjacent. By target reduction, it is possible to extract the Bell pairs only if $\{b_1, b_2\}$ and the isolated vertex a_2 can be extracted from $Z_{a_1}(T')$. $a_2 \sim_F b_1$ in $Z_{a_1}(T')$, so it is not possible to extract the Bell pairs by Lemma 2.2; none of the descriptions correspond to $\{a_2, b_1\}$ in $K_2 \cup K_2$.

Suppose $a_1 > a_2 > b_1$ is not satisfied. WLOG, we have that a_1 is a leaf or axil of b_1 in T' . Extraction of the Bell pairs is impossible; otherwise this would contradict Lemma 2.2, since $a_1 \sim_F b_1$ but none of the descriptions correspond to $\{a_1, b_1\}$ in $K_2 \cup K_2$. $\square$

Lemma 5.5. Let $a_1, a_2, b_1, b_2 > m$ be vertices in $T_{m,n}$. One can extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ using local Clifford operations, Pauli measurements, and classical communication iff $a_1 > a_2 > b_1 > b_2$ (up to some reordering) and a_2 is not adjacent to b_1 .

Proof. Apply source reduction to delete any vertices $v > a_1, a_2, b_1, b_2$ and call the new graph T' . Suppose $a_1 > a_2 > b_1 > b_2$ and a_2 and b_1 are not adjacent. One can extract the Bell pairs by deleting all vertices in between a_2 and b_1 , Y-measuring any vertices in between b_1 and b_2 , and then deleting any remaining vertices not in the Bell pairs.

Suppose $a_1 > a_2 > b_1 > b_2$ but a_2 and b_1 are adjacent. By target reduction, it is possible to extract the Bell pairs only if $\{b_1, b_2\}$ and the isolated vertex a_2 can be extracted from $Z_{a_1}(T')$. $a_2 \sim_F b_1$ in $Z_{a_1}(T')$, so it is not possible to extract the Bell pairs by Lemma 2.2; none of the descriptions correspond to $\{a_2, b_1\}$ in $K_2 \cup K_2$.

Suppose $a_1 > a_2 > b_1 > b_2$ is not satisfied. WLOG, we have that a_1 is a leaf or axil of b_1 in T' . Extraction of the Bell pairs is impossible; otherwise this would contradict Lemma 2.2, since $a_1 \sim_F b_1$ but none of the descriptions correspond to $\{a_1, b_1\}$ in $K_2 \cup K_2$. $\square$

Theorem 5.1. Using local Clifford operations, Pauli measurements, and classical communication, one can extract Bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ from $T_{m,n}$ iff the paths from a_1 to a_2 and from b_1 to b_2 are non-adjacent or one of the following cases is satisfied, up to some reordering:

- $a_1, a_2, b_1, b_2 \leq m$, no 3 of $\{a_1, a_2, b_1, b_2\}$ are consecutive, and $a_1 < a_2 < b_1 < b_2$
- $b_1, b_2 < a_2 = m < a_1$ and a_2, b_1, b_2 are not consecutive in that order
- $a_2, b_1, b_2 < m < a_1$, no 3 of $\{m, a_2, b_1, b_2\}$ are consecutive, and $m > a_2 > b_1 > b_2$

Proof. All cases are covered by lemmas 6.1-6.5. $\square$

5.2 Double Ring Graphs

Definition 5.2. Let the double ring graph $DR_{m,n}(V_{m,n}, E_{m,n})$ satisfy the following:

$$V_{m,n} = \{1, 2, \dots, m+n-2\} \quad (23)$$

$$E_{m,n} = \{\{1, 2\}, \{2, 3\}, \dots, \{m, 1\}\} \cup \{\{m-1, m\}, \{m, m+1\}, \dots, \{m+n-2, m-1\}\}, \quad (24)$$

where $m, n > 2$.

The double ring graph was of interest to us since the 2×5 grid is locally equivalent to a double ring graph. We prove constructions and bottlenecks for cases when vertices $m-1$ and m are included in the Bell pairs to be extracted.

Lemma 5.6. Let $a, b < m-1$ be vertices in $DR_{m,n}$. One can extract Bell pairs $\{a, m-1\}$ and $\{b, m\}$ from $DR_{m,n}$ using local Clifford operations, Pauli measurements, and classical communication iff $a < m-1 < m < b$ (up to some reordering) and either (1) no 3 of $\{a, b, m-1, m\}$ are consecutive or (2) a and b are not adjacent.

Proof. If no 3 of $\{a, b, m-1, m\}$ are consecutive and $a < m-1 < m < b$ (up to some reordering), one can delete every vertex $v > m$ to obtain R_m . Then extracting the Bell pairs is possible by Theorem V.3.

If $a < m-1 < m < b$ (up to some reordering) and a and b are not adjacent, delete all vertices between a and b and Y-measure every vertex $v > m$ to extract the Bell pairs.

If $n = 3$, $G' = \tau_{m+1}(DR_{m,n})$ is a ring. If $b < m-1 < m < a$ in $DR_{m,n}$, $b < m-1 < m < a$ in G' , so it is impossible to extract the Bell pairs by Theorem 2.5. If $b, a, m-1, m$

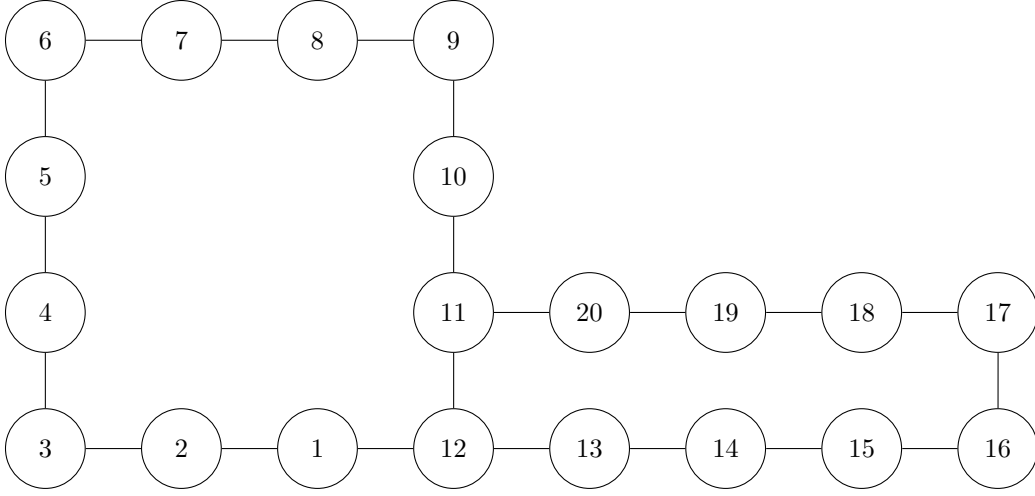


Figure 20: $DR_{12,10}$

were consecutive in $DR_{m,n}$, then $a, b, m-1$ or are consecutive in G' , so it is impossible to extract the Bell pairs by Theorem 2.5.

Take any graph $DR_{m,n}$ with vertices $a, b < m-1$. Suppose that $b < m-1 < m < a$ or that $b, a, m-1, m$ are consecutive. We prove that it is not possible to extract the Bell pairs by induction on the number of vertices in $LB_{m,n}$.

Base case: Suppose $DR_{m,n}$ has 5 vertices. Then $n = 3$, so the lemma holds.

Inductive step: Assume that the lemma holds for large butterfly graphs on 5 up to k vertices. Let $DR_{m,n}$ be on $k+1$ vertices and have $b < m-1 < m < a$ or $b, a, m-1, m$ are consecutive, in that order.

If $n = 3$, we already showed that the lemma holds. Assume $n > 3$. By the contrapositive of Theorem 2.1, if it is possible to extract the Bell pairs from $DR_{m,n}$, then there exists some vertex v such that it is possible to extract the Bell pairs from at least one of (1) $Z_v(DR_{m,n})$, (2) $Y_v(DR_{m,n})$, or (3) $X_v(DR_{m,n})$. We prove that it is not possible to extract the Bell pairs from any of these new graphs, and therefore that the lemma holds for double ring graphs on $k+1$ vertices.

Case 1: Suppose $v > m$. Repeatedly apply source reduction to delete every vertex $u > m$ in $Z_v(DR_{m,n})$, resulting in R_m . By Theorem 2.5, it is not possible to extract the Bell pairs from R_1 since 3 of $a, b, m-1, m$ are consecutive or $a < m-1 < m < b$ is not satisfied.

Suppose $b, a, m-1, m$ are consecutive, in that order, and $v < m-1$. For each vertex u that was in between v and m in $DR_{m,n}$, apply source reduction to delete u in $Z_v(DR_{m,n})$. $Z_v(DR_{m,n})$ becomes a tadpole graph. It is not possible to extract the Bell pairs by Theorem 6.1.

Suppose $b < m-1 < m < a$. Repeatedly apply source reduction to delete leaves and

axils in $Z_v(DR_{m,n})$ until either $a \sim_F m$, $a \sim_F b$, or $m-1 \sim_F b$. It is not possible to extract the Bell pairs; otherwise, this would contradict Lemma 2.2 since none of the descriptions match $\{a, m\}$, $\{a, b\}$, or $\{m-1, b\}$ in $K_2 \cup K_2$.

Case 2: The Bell pairs cannot be extracted from $Y_v(DR_{m,n})$ by the inductive hypothesis, since $Y_v(DR_{m,n})$ has a large butterfly graph on k vertices. The ordering of $a, b, m-1, m$ is preserved, and, if any 2 vertices of the Bell pairs were adjacent in $DR_{m,n}$, they are also adjacent in $Y_v(DR_{m,n})$.

Case 3: Suppose v is adjacent to some vertex $u \neq x, y, a, b$. Then $X_v(DR_{m,n})$ is a large butterfly graph or a ring on $k-1$ vertices plus vertex u , which is now a leaf. Apply source reduction to delete u . By the inductive hypothesis or Theorem 2.5, it is impossible to extract the Bell pairs from $Z_u \circ X_v(DR_{m,n})$, since the ordering of vertices $m-1, m, a, b$ is unchanged and if any 2 of them were adjacent in $DR_{m,n}$, they are also adjacent in $Z_u \circ X_v(DR_{m,n})$.

Suppose $b, a, m-1, m$ are consecutive, in that order, and v is adjacent to b and m . Then $b \sim_F m$ in $X_v(DR_{m,n})$. WLOG, assume m is a leaf of b . By target reduction, it is possible to extract the Bell pairs from $X_v(DR_{m,n})$ only if it is possible to extract $\{a, m-1\}$ and the isolated vertex b from $Z_m \circ X_v(DR_{m,n})$. However, this is not possible by Lemma 2.3 since $\tau_b \circ Z_m \circ X_v(DR_{m,n})$ is a ring graph with $b, a, m-1$ consecutive, in that order.

Suppose $b < m-1 < m < a$ and v neighbors both b and $m-1$, both a and m , or both a and b . Then $b \sim_F m-1$, $a \sim_F m$, or $b \sim_F a$, respectively. It is impossible to extract the Bell pairs; otherwise this would contradict Lemma 2.1 since none of the descriptions correspond to $\{b, m-1\}$, $\{a, m\}$, or $\{b, a\}$ in $K_2 \cup K_2$. $\square$

Lemma 5.7. Let $a, b < m$ be vertices in $DR_{m,n}$. One can extract Bell pairs $\{a, b\}$ and $\{m-1, m\}$ from $DR_{m,n}$ using local Clifford operations, Pauli measurements, and classical communication iff no 3 of $\{a, b, m-1, m\}$ are consecutive or a and b are not adjacent.

Proof. If no 3 of $\{a, b, m-1, m\}$ are consecutive. One can delete all vertices $v > m$ to obtain R_m . Extracting the Bell pairs is possible by Theorem 2.5, since $a < b < m-1 < m$ or $b < a < m-1 < m$.

If a and b are not adjacent then Y-measure any vertices $v > m+1$. Next, perform a local complementation on $m+1$ to get a ring where $a < b < m-1 < m$ or $b < a < m-1 < m$ and no 3 of $\{a, b, m-1, m\}$ are consecutive. Extracting the Bell pairs is possible by Theorem 2.5.

If $n = 3$, $\tau_{m+1}(DR_{m,n})$ is a ring graph on $m+1$ vertices. If a and b are adjacent and 3 of $\{a, b, m-1, m\}$ are consecutive in $DR_{m,n}$, assume WLOG that $a, b, m-1$ are consecutive, in that order. $a, b, m-1$ are still consecutive in $\tau_v(LB_{m,n})$, so the Bell pairs cannot be extracted by Theorem 2.5.

Take any graph $DR_{m,n}$ where $a, b < m-1$ are vertices in $DR_{m,n}$. Suppose that $a, b, m-1, m$ are consecutive, in that order. We prove that it is not possible to extract the Bell pairs by induction on the number of vertices in $DR_{m,n}$.

Base case: Suppose $LB_{m,n}$ has 5 vertices. Then $n = 3$, so the lemma holds.

Inductive step: Assume that the lemma holds for double ring graphs on 5 up to k vertices. Let $DR_{m,n}$ be on $k+1$ vertices and have that $a, b, m-1, m$ are consecutive, in that order. If $n = 3$, we already showed that the lemma holds, so assume $n > 3$. By the contrapositive of Theorem 2.1, if it is possible to extract the Bell pairs from $DR_{m,n}$, then, for some vertex v , it is possible to extract the Bell pairs from at least one of (1) $Z_v(DR_{m,n})$, (2) $Y_v(DR_{m,n})$, or (3) $X_v(DR_{m,n})$. We prove that it is not possible to extract the Bell pairs from any of these new graphs, and therefore that the lemma holds for double ring graphs on $k+1$ vertices.

Case 1: If $v > m$, then repeatedly apply source reduction to delete every vertex $u > m$ in $Z_v(LB_{m,n})$. This yields R_m . By Theorem 2.5, it is not possible to extract the Bell pairs, since 3 of $\{a, b, m-1, m\}$ are consecutive.

If $v < m-1$, repeatedly apply source reduction to delete every vertex $u < a$ in $Z_v(DR_{m,n})$ to obtain a tadpole graph. It is not possible to extract the Bell pairs by Theorem 6.1.

Case 2: The Bell pairs cannot be extracted from $Y_v(DR_{m,n})$ by the inductive hypothesis, since $Y_v(DR_{m,n})$ is a double ring graph on k vertices. $a, b, m-1, m$ are still consecutive in $Y_v(DR_{m,n})$.

Case 3: If v is adjacent to some vertex $u \neq a, b, m-1, m$, then $X_v(DR_{m,n})$ is a double ring graph on $k-1$ vertices plus vertex u , which is now a leaf. Apply source reduction to delete u . By the inductive hypothesis or Theorem 2.5, it is impossible to extract the Bell pairs from $Z_u \circ X_v(DR_{m,n})$, since $a, b, m-1, m$ are still consecutive in $Z_u \circ X_v(DR_{m,n})$.

Otherwise, v neighbors both a and m . Then $a \sim_F m$ in $X_v(DR_{m,n})$. It is impossible to extract the Bell pairs; otherwise this would contradict Lemma 2.2 since none of the descriptions correspond to $\{a, m\}$ in $K_2 \cup K_2$. $\square$

Lemma 5.8. Let $a < m-1$ and $b > m$ be a vertices in $DR_{m,n}$. One can extract Bell pairs $\{a, m-1\}$ and $\{b, m\}$ from $DR_{m,n}$ using local Clifford operations, Pauli measurements, and classical communication iff $a, b, m-1, m$ are not all consecutive in some order and $m, n > 3$.

Proof. Suppose $a, b, m-1, m$ are not all consecutive in some order and $m, n > 3$. If a is adjacent to m or $m-1$, Y-measure any vertices in between b and m or $m-1$, respectively. If b is adjacent to m or $m-1$, Y-measure any vertices in between a and m or $m-1$, respectively. Otherwise, Y-measure the vertices in between a and m and the vertices in between b and m . WLOG, assume a and b are adjacent to m but not $m-1$. Y-measure any vertices v satisfying $a < v < m-2$. Call the resulting graph G_1 . Then $\tau_{m-2} \circ \tau_a \circ \tau_{m-2}(G_1)$ is a ring graph where no 3 of $\{a, m-1, m, b\}$ are consecutive and $a < m-1 < m < b$. By Theorem 2.5, it is possible to extract the Bell pairs.

If m or $n = 3$, then it is impossible to extract the Bell pairs; WLOG let $m = 3$. Then $\tau_a(DR_{m,n})$ is a ring where $m-1, a, m$ are consecutive. By Theorem 2.5, it is impossible to extract the Bell pairs.

If m or $n = 4$, then it is also impossible to extract the Bell pairs; WLOG let $m = 4$. Let $G' = \tau_1 \circ \tau_2 \circ \tau_1(DR_{m,n})$, a ring graph. If $a, m-1, m, b$ were consecutive, in that order, in $DR_{m,n}$, then a, m, b are consecutive in G' . If $b, m-1, m, a$ were consecutive, in that order, in $DR_{m,n}$, then $a, m-1, b$ are consecutive in G' . By Theorem 2.5, it is impossible to extract the Bell pairs.

Suppose either $a, m-1, m, b$ or $b, m-1, m, a$ are consecutive, in that order. We prove that it is not possible to extract the Bell pairs by induction on the number of vertices in $DR_{m,n}$.

Bases cases: If $DR_{m,n}$ is on 4 or 5 vertices, then m or $n = 3$, so it is impossible to extract the Bell pairs.

Inductive step: Assume that the lemma holds for double ring graphs on 4 up to k vertices. Let $DR_{m,n}$ be on $k+1$ vertices and have that either $a, m-1, m, b$ or $b, m-1, m, a$ are consecutive, in that order.

We already proved it is impossible to extract the Bell pairs if m or n equals 3 or 4. Assume $m, n > 4$. By the contrapositive of Theorem 2.1, if it is possible to extract the Bell pairs from $DR_{m,n}$, then, for some vertex v , it is possible to extract the Bell pairs from at least one of (1) $Z_v(DR_{m,n})$, (2) $Y_v(DR_{m,n})$, or (3) $X_v(DR_{m,n})$. We prove that it is not possible to extract the Bell pairs from any of these new graphs, and therefore that the lemma holds for double ring graphs on $k+1$ vertices.

Case 1: Repeatedly apply source reduction to delete every leaf $u \neq a, b$ in $Z_v(LB_{m,n})$ until no such leaf exists. One obtains a tadpole graph with $a, m-1, m, b$ or $b, m-1, m, a$ still consecutive, in that order. By Theorem 6.1, it is impossible to extract the Bell pairs.

Case 2: The Bell pairs cannot be extracted from $Y_v(DR_{m,n})$ by the inductive hypothesis, since $Y_v(DR_{m,n})$ is a double ring graph on k vertices where $a, m-1, m, b$ or $b, m-1, m, a$ are still consecutive, in that order.

Case 3: Since $m, n > 4$ and $a, m-1, m, b$ or $b, m-1, m, a$ are consecutive, v is adjacent to some vertex $u \neq m-1, m, a, b$. $X_v(DR_{m,n})$ is a double ring graph on $k-1$ vertices plus vertex u , which is now a leaf. Apply source reduction to delete u . By the inductive hypothesis, it is impossible to extract the Bell pairs, since $a, m-1, m, b$ or $b, m-1, m, a$ are still consecutive, in that order. $\square$

Lemma 5.9. Let $a < m-1$ and $b > m$ be a vertices in $DR_{m,n}$. It is not possible to extract Bell pairs $\{a, b\}$ and $\{m-1, m\}$ from $DR_{m,n}$ using local Clifford operations, Pauli measurements, and classical communication.

Proof. If m or $n = 3$, then it is impossible to extract the Bell pairs: WLOG let $m = 3$. Then $\tau_a(LB_{m,n})$ is a ring where $a < b < m-1 < m$ is not satisfied. By Theorem 2.5, it is impossible to extract the Bell pairs.

We prove the lemma by induction on the number of vertices in $DR_{m,n}$.

Base cases: If $DR_{m,n}$ is on 4 or 5 vertices, then m or $n = 3$, so it is impossible to extract the Bell pairs.

Inductive step: Assume that the lemma holds for double ring graphs on 4 up to k vertices. Let $DR_{m,n}$ be on $k + 1$ vertices.

By the contrapositive of Theorem 2.1, if it is possible to extract the Bell pairs from $DR_{m,n}$, then, for some vertex v , it is possible to extract the Bell pairs from at least one of (1) $Z_v(DR_{m,n})$, (2) $Y_v(DR_{m,n})$, or (3) $X_v(DR_{m,n})$. We prove that it is not possible to extract the Bell pairs from any of these new graphs, and therefore that the lemma holds for double ring graphs on $k + 1$ vertices.

Case 1: Repeatedly apply source reduction to delete every leaf and then every axil $u \neq a, b, m - 1, m$ in $Z_v(DR_{m,n})$ until no such leaf or axil exists. In the resulting graph, $a \sim_F m$, $a \sim_F m - 1$, $b \sim_F m$, or $a \sim_F m - 1$. It is impossible to extract the Bell pairs; otherwise this would contradict Lemma 2.2 since none of the descriptions correspond to $\{a, m\}$, $\{a, m - 1\}$, $\{b, m\}$, or $\{b, m - 1\}$ in $K_2 \cup K_2$.

Case 2: The Bell pairs cannot be extracted from $Y_v(DR_{m,n})$ by the inductive hypothesis, since $Y_v(DR_{m,n})$ is a double ring graph on k vertices where $a < m - 1$ and $b > m$ still.

Case 3: If v is adjacent to some vertex $u \neq m - 1, m, a, b$, then $X_v(DR_{m,n})$ is a double ring graph on $k - 1$ vertices plus vertex u , which is now a leaf. Apply source reduction to delete u . By the inductive hypothesis, it is impossible to extract the Bell pairs, since $a < m - 1$ and $b > m$ still.

Otherwise, WLOG, v neighbors both a and m . Then $a \sim_F m$ in $X_v(DR_{m,n})$. It is impossible to extract the Bell pairs; otherwise this would contradict Lemma 2.2 since none of the descriptions correspond to $\{a, m\}$ in $K_2 \cup K_2$. $\square$

6 Conclusion

In this write-up, we proved constructions and bottlenecks for $2 \times n$ and $3 \times n$ grids, in addition to tadpole and double ring graphs.

It may be possible to extend protocols for extracting Bell pairs from $3 \times n$ grids to be used for $m \times n$ grids. Further research is needed to completely solve grids and to resolve the other cases in double ring graphs.

References

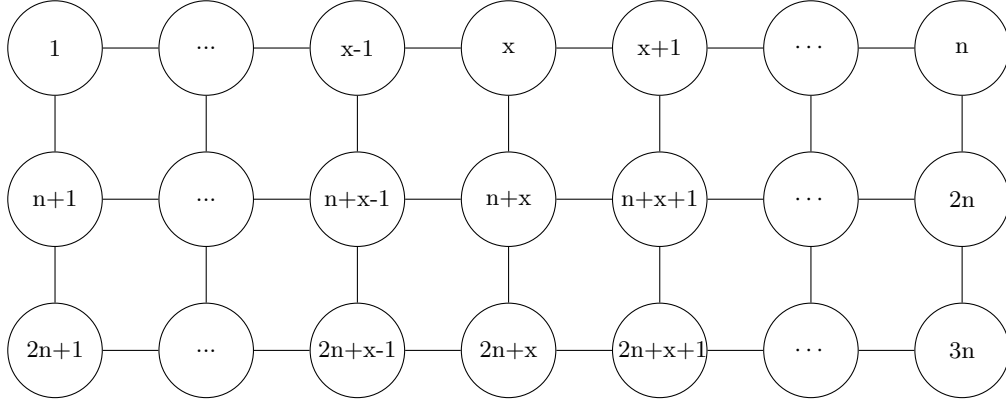
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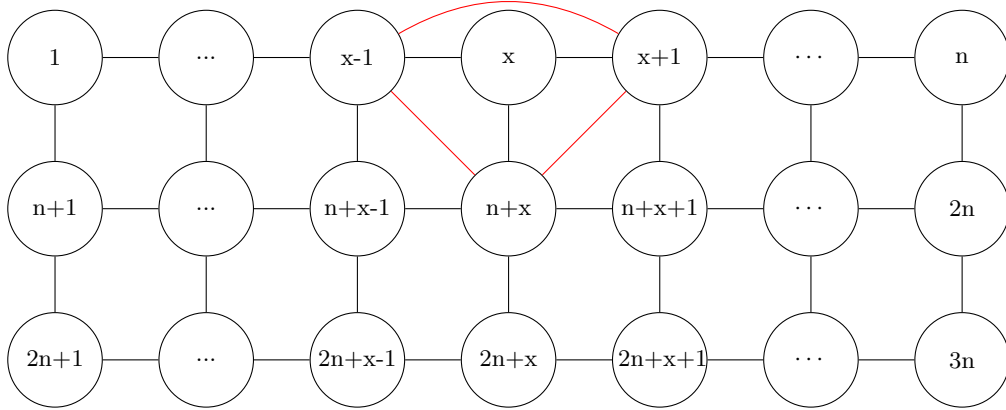
Appendices

A Intermediary Graphs for Auxiliary Protocols

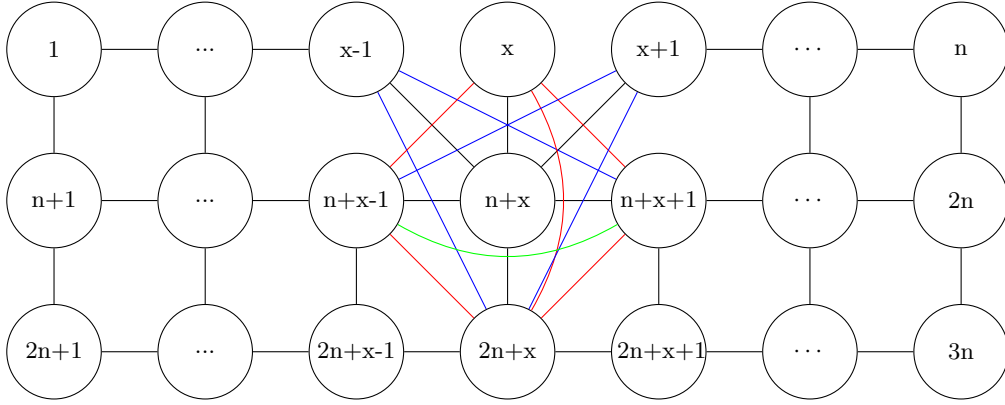
A.1 Crossing Protocol - Definition 2.10



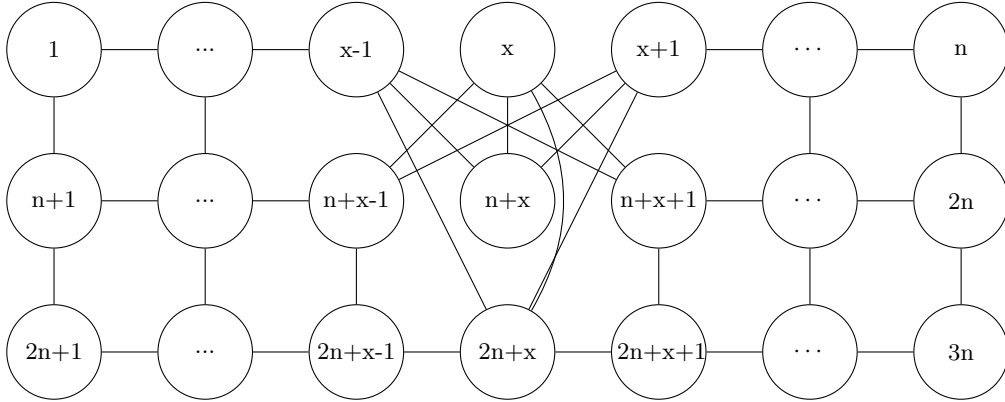
(a) G is a $3 \times n$ grid where $n \geq 3$. We will perform the crossing protocol on G to get $G_5 = \chi_{n+x}(G)$.



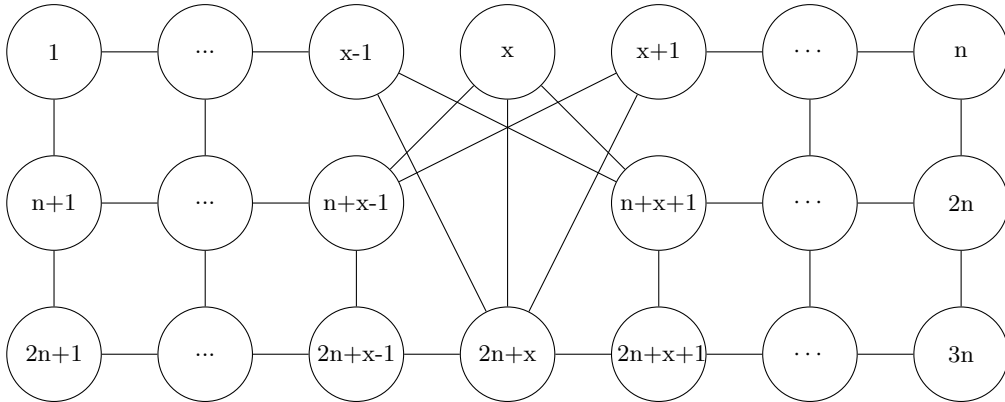
(b) G_1 , the first intermediary graph. Perform a local complementation on vertex x to get $G_1 = \tau_x(G)$. Added edges from the LC are colored.



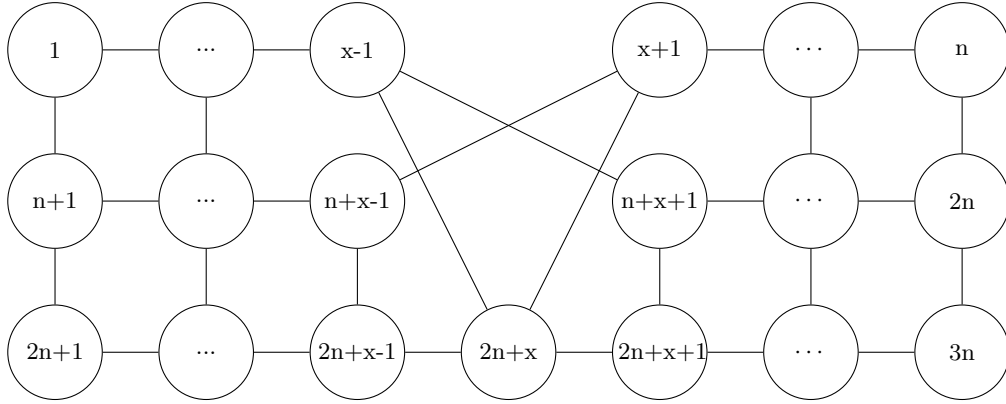
(c) G_2 , the second intermediary graph. Perform a local complementation on vertex $n+x$ to get $G_2 = \tau_{n+x}(G_1)$. Added edges from the LC are colored.



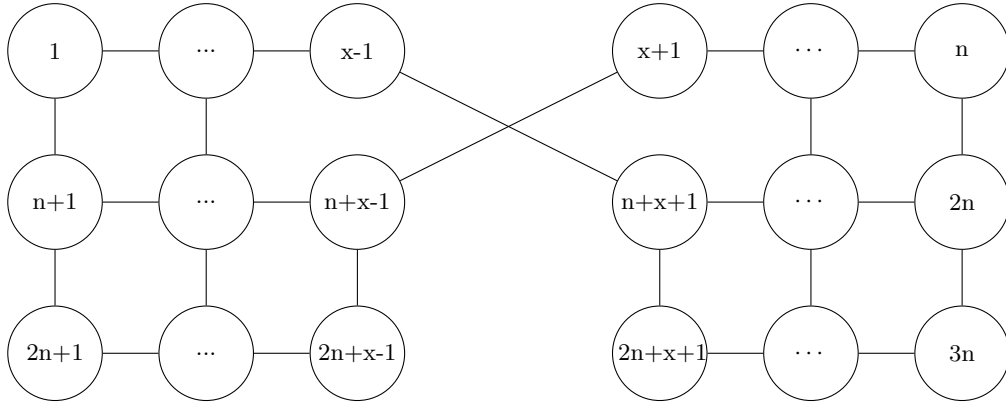
(d) G_3 , the third intermediary graph. Perform a local complementation on vertex x to get $G_3 = \tau_x(G_2)$. Added edges from the LC are colored.



(e) G_4 , the fourth intermediary graph. Perform a vertex deletion on vertex $n+x$ to get $G_4 = G_3 \setminus V_{n+x}$.



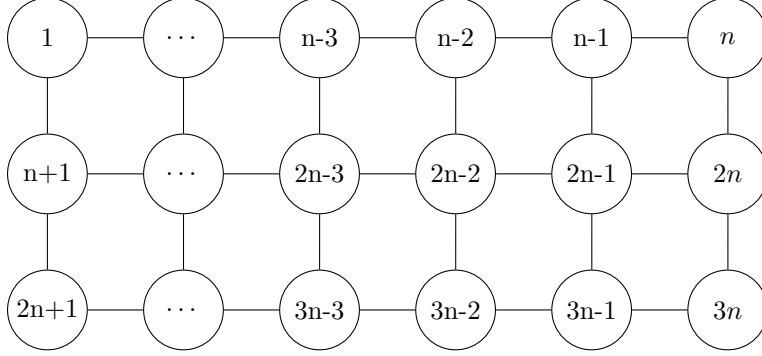
(f) G_5 , the fifth intermediary graph. Perform a vertex deletion on vertex x to get $G_5 = G_4 \setminus V_x$.



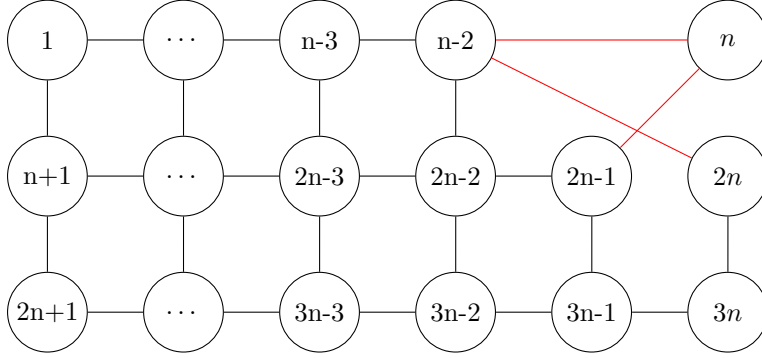
(g) G' , the resulting graph. Perform a vertex deletion on vertex $2n+x$ to get $G' = G_5 \setminus V_{2n+x} = \chi_{n+x}(G)$.

Figure 21

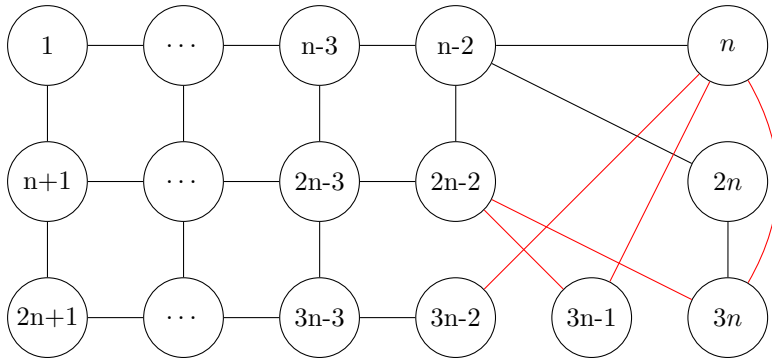
A.2 Flipping Protocol - Definition 2.11



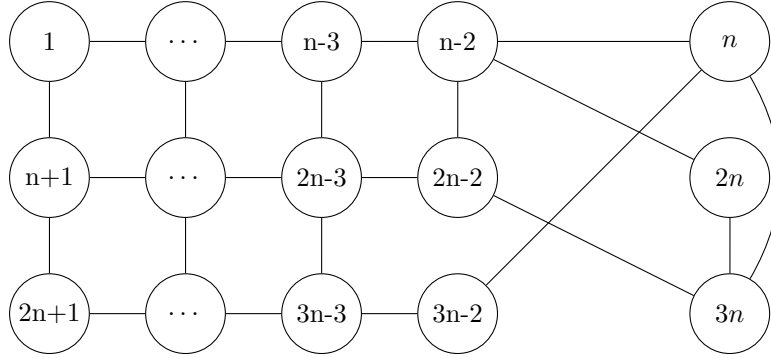
(a) G is a $3 \times n$ grid where $n \geq 4$. We will perform the flipping protocol on G to get $G' = \varphi(G)$.



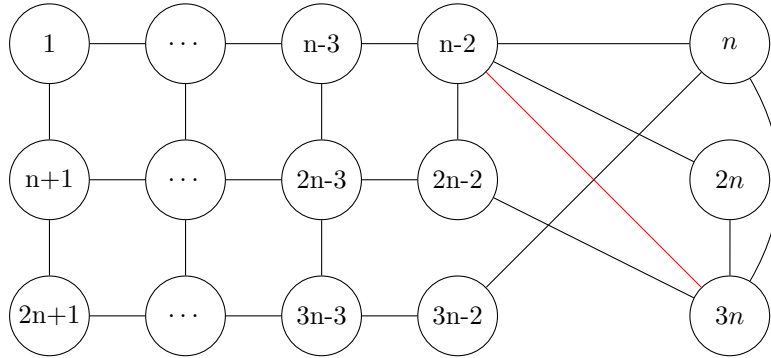
(b) G_1 , the first intermediary graph. Perform a X measurement on vertex $\{n-1\}$, using $\{n\}$ as a pivot, to get $G_1 = X_{n-1}^n(G)$. Added edges from the LCs are colored.



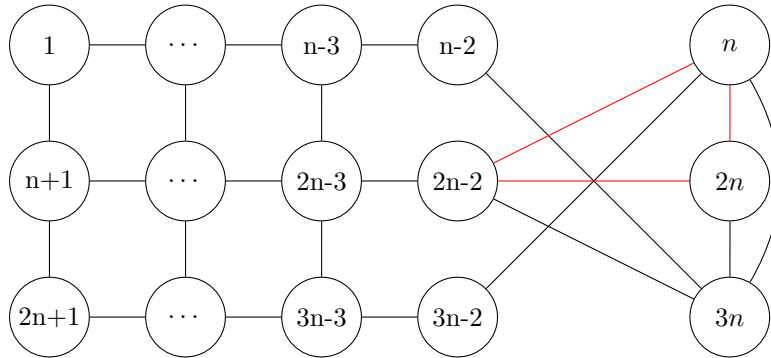
(c) G_2 , the second intermediary graph. Perform a X measurement on vertex $\{2n-1\}$, using $\{3n-1\}$ as a pivot, to get $G_2 = X_{2n-1}^{3n-1}(G_1)$. Added edges from the LCs are colored.



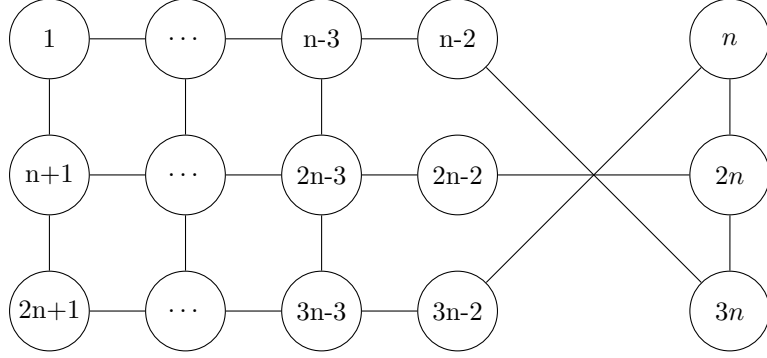
(d) G_3 , the third intermediary graph. Perform a Z measurement on vertex $\{3n-1\}$ to get $G_3 = Z_{3n-1}(G_2)$.



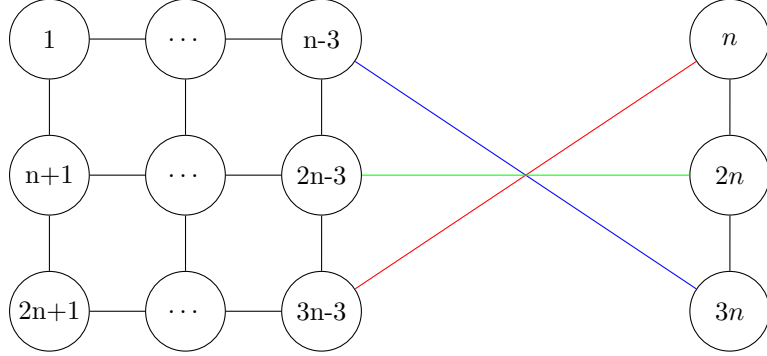
(e) G_4 , the fourth intermediary graph. Perform a local complementation on vertex $\{2n\}$ to get $G_4 = \tau_{2n}(G_3)$. Added edges from the LCs are colored.



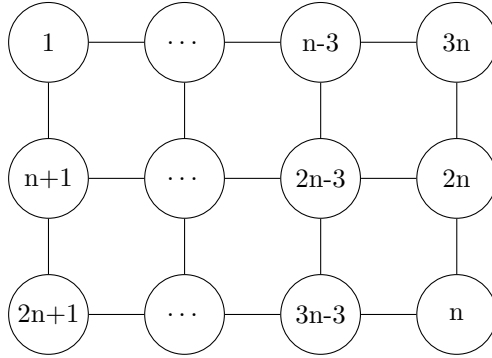
(f) G_5 , the fifth intermediary graph. Perform a local complementation on vertex $\{3n\}$ to get $G_5 = \tau_{3n}(G_4)$. Added edges from the LCs are colored.



(g) G_6 , the sixth intermediary graph. Perform a local complementation on vertex $\{2n\}$ to get $G_6 = \tau_{2n}(G_5)$.



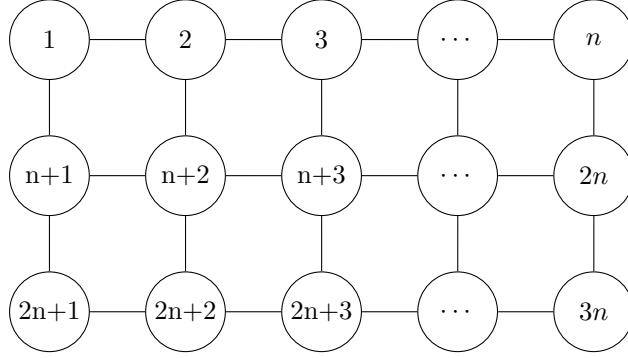
(h) G' , the resulting graph. Perform Y measurements on vertices $\{n-2, 2n-2, 3n-2\}$ to get $G_6 = Y_{3n-2} \circ Y_{2n-2} \circ Y_{n-2}(G_5)$.



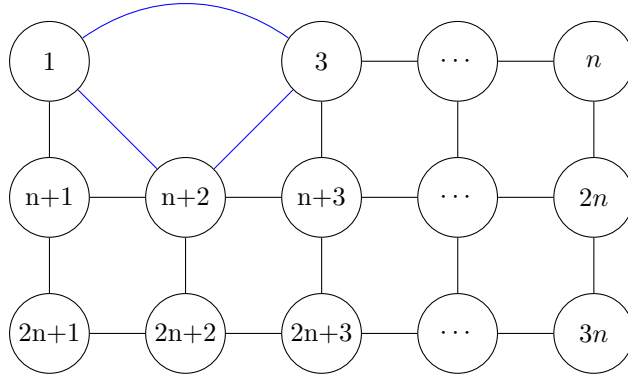
(i) G' , the resulting graph. $G' = \varphi(G)$

Figure 22

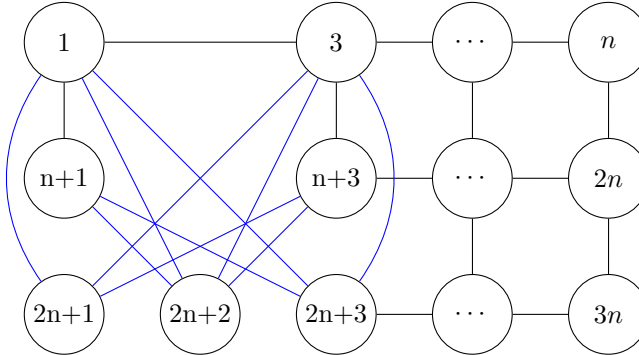
A.3 Corner Extraction Protocol - Definition 2.12



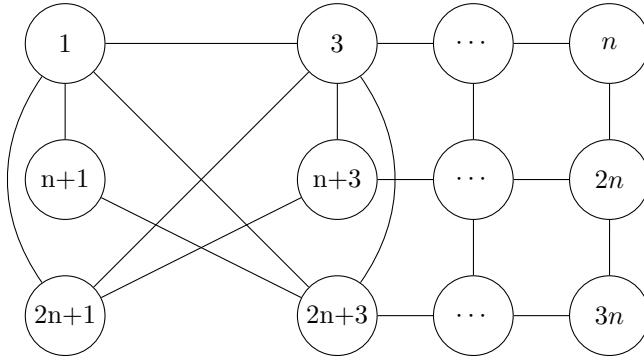
(a) G is a $3 \times n$ grid where $n \geq 3$. We will perform the Corner Extraction protocol on G to get $G' = \kappa(G)$.



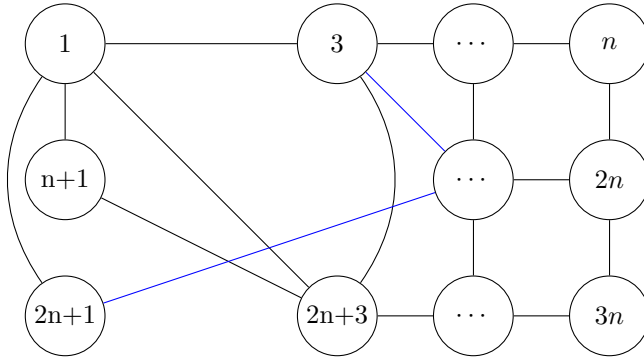
(b) G_1 , the first intermediary graph. Perform a Y measurement on vertex $\{2\}$ to get $G_1 = Y_2(G)$. Added edges from the LCs are colored.



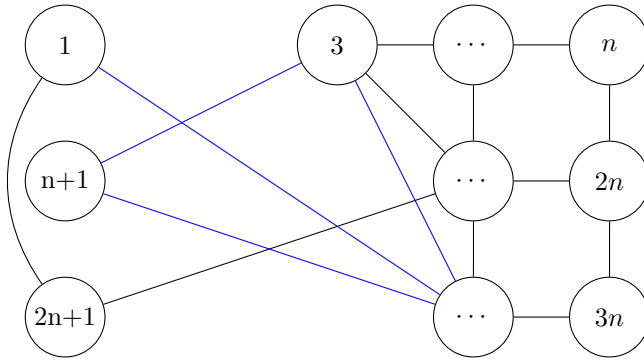
(c) G_2 , the second intermediary graph. Perform a X measurement on vertex $\{n+2\}$, using $\{2n+2\}$ as a pivot, to get $G_2 = X_{n+2}^{2n+2}(G_1)$. Added edges from the LCs are colored.



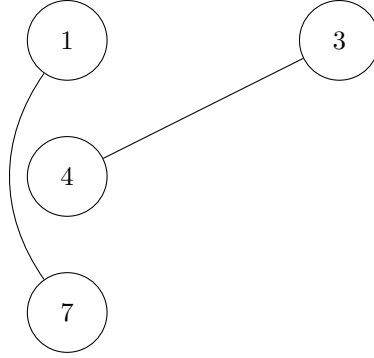
(d) G_3 , the third intermediary graph. Perform a Z measurement on vertex $\{2n+2\}$ to get $G_3 = Z_{2n+2}(G_2)$.



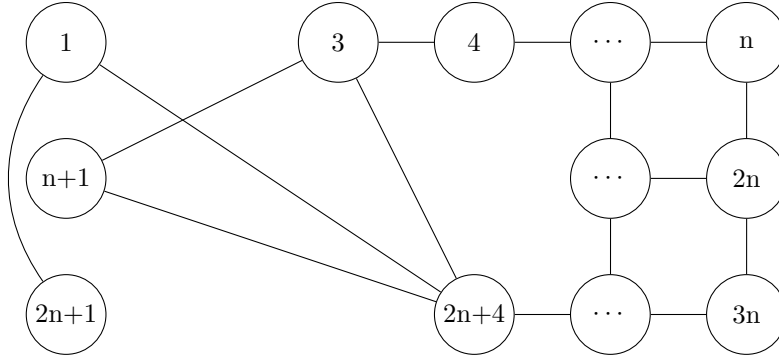
(e) G_4 , the fourth intermediary graph. Perform a Y measurement on vertex $\{n+3\}$ to get $G_4 = Y_{n+3}(G_3)$. Added edges from the LCs are colored.



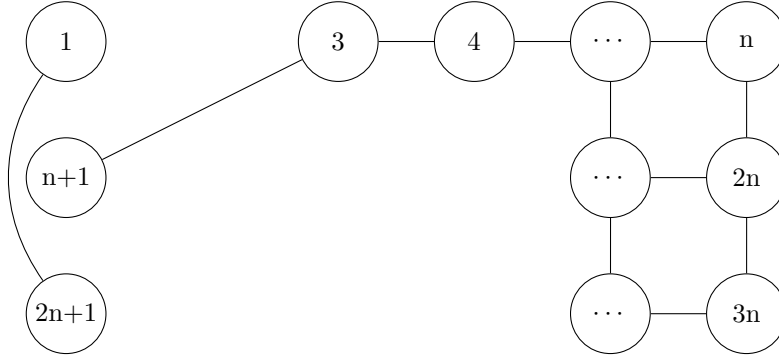
(f) G_5 , the fifth intermediary graph. Perform a Y measurement on vertex $\{2n+3\}$ to get $G_5 = Y_{2n+3}(G_4)$. Added edges from the LCs are colored.



(g) If $n = 3$, $G' = G_5 = \kappa(G)$. G' the resulting graph.



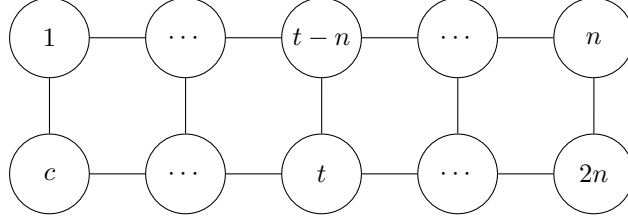
(h) If $n \geq 4$, G_6 , the sixth intermediary graph. Perform a Z measurement on vertex $\{n + 4\}$ to get $G_6 = Z_{n+4}(G_5)$.



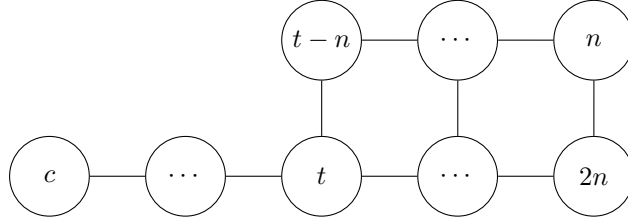
(i) If $n \geq 4$, G' , the resulting graph. $G' = \kappa(G)$. Perform a Z measurement on vertex $\{2n + 4\}$ to get $G' = G_7 = Z_{2n+4}(G_6)$.

Figure 23

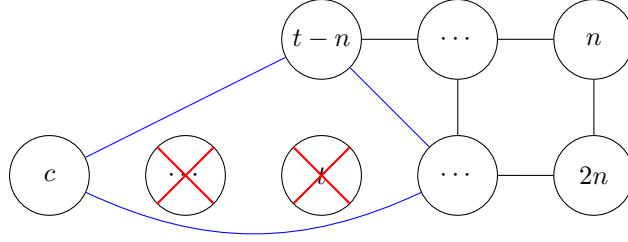
A.4 Corner Replacement Protocol - Definition 2.13



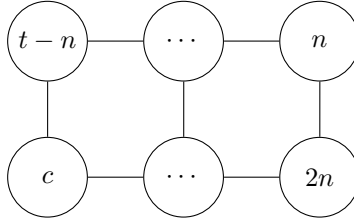
(a) G is a $2 \times n$ grid where $n \geq 3$. We will perform the Corner Replacement Protocol, ρ_c^t , to get $G' = \rho_c^t(G)$. For simplicity we will just show the case where $c = n + 1$.



(b) G_1 , the first intermediary graph. Perform Z measurements on all vertices between $(1, t-n)$, as well as vertex $\{1\}$, to get $G_1 = Z_{t-n-1} \circ \dots \circ Z_2 \circ Z_1(G)$.



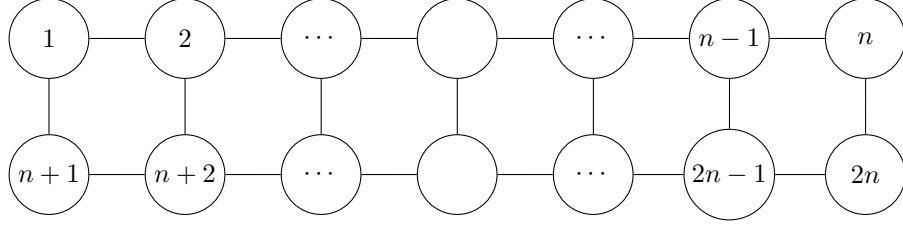
(c) G_2 , the second intermediary graph. Perform Y measurements on all vertices between $[c+1, t]$, inclusive, to get $G_2 = Y_t \circ \dots \circ Y_{c+2} \circ Y_{c+1}(G)$. Added edges from the LC's are colored.



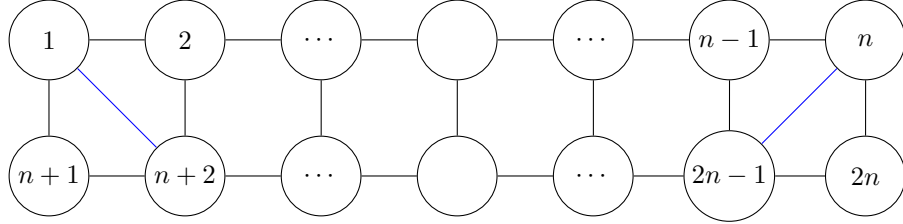
(d) $G' = \rho_c^t(G)$, the resulting graph. Perform a local complementation on vertex $\{c\}$ to get $G' = G_3 = \tau_c(G_2)$.

Figure 24

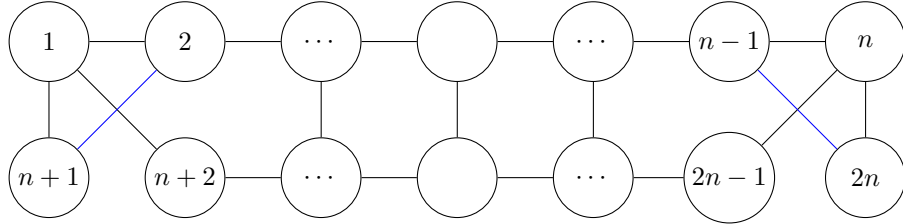
A.5 Corner Folding Protocol - Definition 2.14



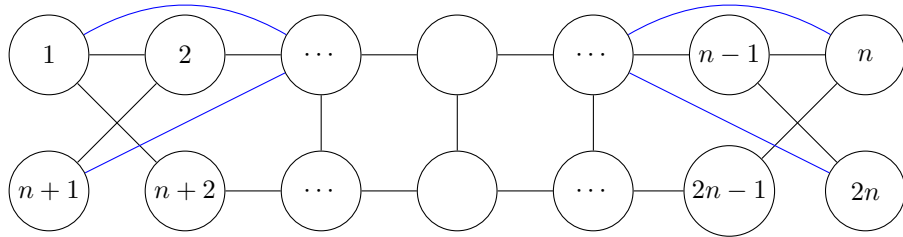
(a) G is a $2 \times n$ grid where $n \geq 5$. We will perform the Corner Folding Protocol, ξ , to get $G' = \xi(G)$.



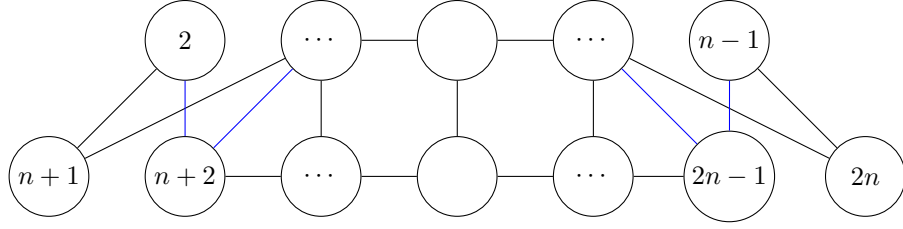
(b) G_1 , the first intermediary graph. Perform local complementations on vertices $\{n+1, 2n\}$, to get $G_1 = \tau_{2n} \circ \tau_{n+1}(G)$. Added edges from the LC's are colored.



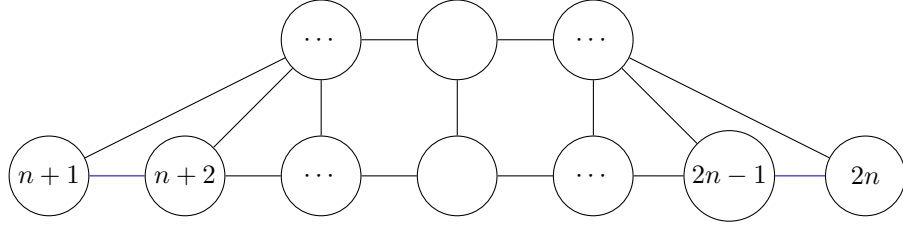
(c) G_2 , the second intermediary graph. Perform local complementations on vertices $\{1, n\}$, to get $G_2 = \tau_n \circ \tau_1(G_1)$. Added edges from the LC's are colored.



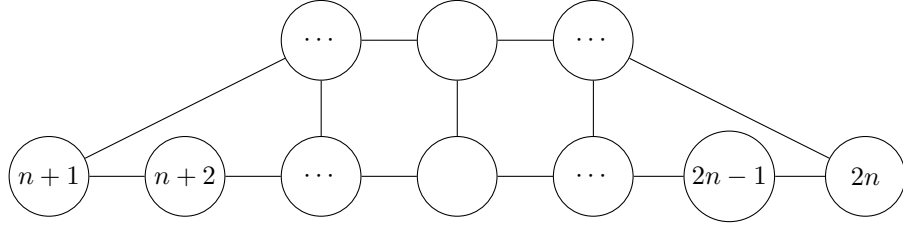
(d) G_3 , the third intermediary graph. Perform local complementations on vertices $\{2, n-1\}$, to get $G_3 = \tau_{n-1} \circ \tau_2(G_2)$. Added edges from the LC's are colored.



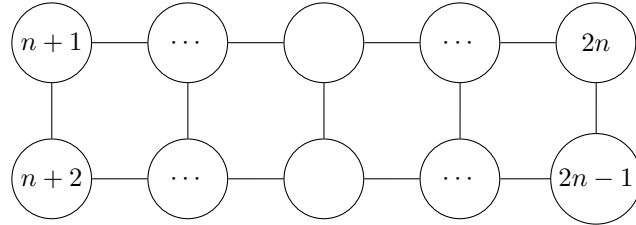
(e) G_4 , the fourth intermediary graph. Perform Y measurement on vertices $\{1, n\}$, to get $G_4 = Y_n \circ Y_1(G_3)$. Added edges from the LC's are colored.



(f) G_5 , the fifth intermediary graph. Perform Y measurement on vertices $\{2, n-1\}$, to get $G_5 = Y_{n-1} \circ Y_2(G_4)$. Added edges from the LC's are colored.



(g) G' , the resulting graph. Perform local complementations on vertices $\{n+1, 2n\}$, to get $G' = G_6 = \tau_{2n} \circ \tau_{n+1}(G_5)$. Added edges from the LC's are colored.



(h) G' is a $2 \times (n-2)$ grid with corner vertices $V_c \in \{n+1, n+2, 2n, 2n-1\}$.

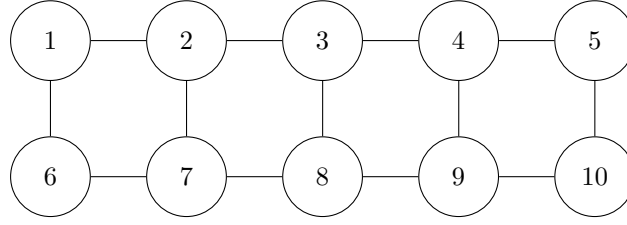
Figure 25

B The 2×5 Grid

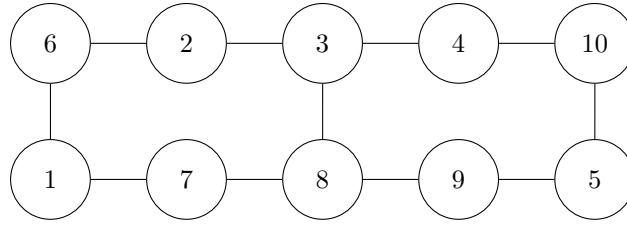
2×5 grids are LC-equivalent to double ring graphs: if G is a 2×5 grid, then

$$G' = \tau_1 \circ \tau_6 \circ \tau_1 \circ \tau_{10} \circ \tau_5 \circ \tau_{10}(G) \quad (25)$$

is a double ring.



(a) G



(b) G'

Figure 26

Applying lemmas 6.6-6.9, we have the following tables indicating whether or not it is possible to extract certain Bell pairs from the 2×5 grid:

		b							
		1	2	4	5	6	7	9	10
a	1	■	×	✓	✓	×	×	✓	✓
	2	✓	■	✓	✓	×	✓	×	✓
	4	✓	✓	■	✓	✓	×	✓	×
	5	✓	✓	×	■	✓	✓	×	×
	6	✓	×	✓	✓	■	✓	✓	✓
	7	×	×	×	✓	×	■	✓	✓
	9	✓	×	×	×	✓	✓	■	×
	10	✓	✓	×	✓	✓	✓	✓	■

(a) Extracting Bell pairs $\{a, 3\}, \{b, 8\}$

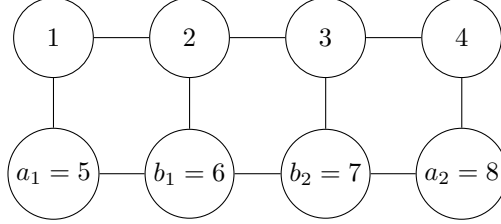
		<i>b</i>							
		1	2	4	5	6	7	9	10
<i>a</i>	1	■	✓	✗	✗	✓	✗	✗	✗
	2	✓	■	✗	✗	✗	✓	✗	✗
	4	✗	✗	■	✓	✗	✗	✓	✗
	5	✗	✗	✓	■	✗	✗	✗	✓
	6	✓	✗	✗	✗	■	✓	✗	✗
	7	✗	✓	✗	✗	✓	■	✗	✗
	9	✗	✗	✓	✗	✗	✗	■	✓
	10	✗	✗	✗	✓	✗	✗	✓	■

(b) Extracting Bell pairs $\{a, b\}, \{3, 8\}$

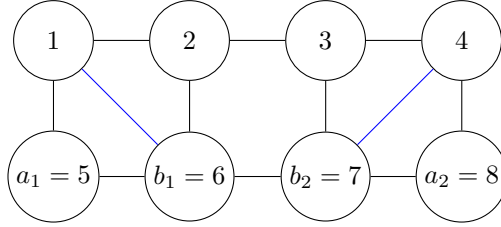
Figure 27

C Intermediary Graphs for Section 3.4

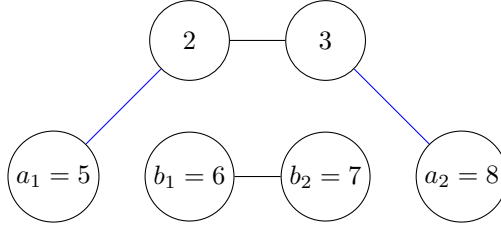
C.1 Intermediary Graphs for Lemma 3.15



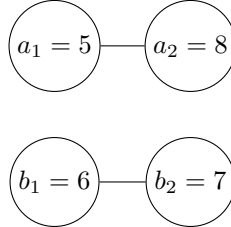
(a) Base Case: G is a 2×4 grid where $5 \leq a_1 < b_1 < b_2 < a_2 \leq 8$. We will simultaneously extract $\{a_1, a_2\}$ $\{b_1, b_2\}$.



(b) Base Case: G_1 , the first intermediary graph. Perform local complementations on vertices $\{5, 8\}$ to get $G_1 = \tau_5 \circ \tau_8(G)$. Added edges from the LCs are colored.

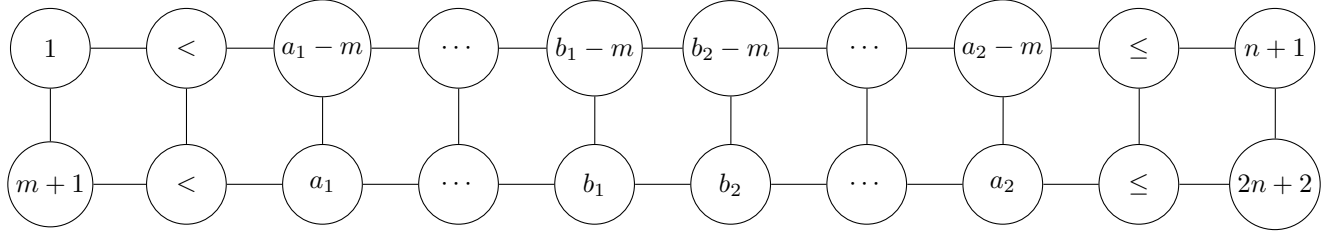


(c) Base Case: G_2 , the second intermediary graph. Perform Y measurements on vertices $\{1, 4\}$ to get $G_2 = Y_1 \circ Y_4(G_1)$. Added edges from the LCs are colored.

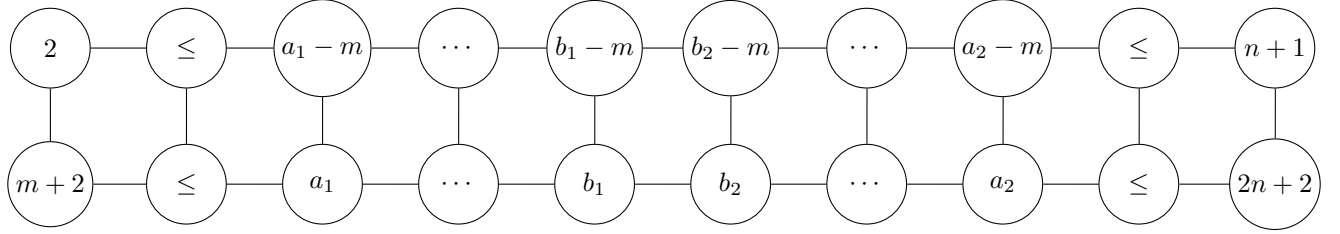


(d) Base Case: G' , the resulting graph containing the extracted bell pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$. Perform Y measurements on vertices $\{2, 3\}$ to get $G' = Y_2 \circ Y_3(G_2)$.

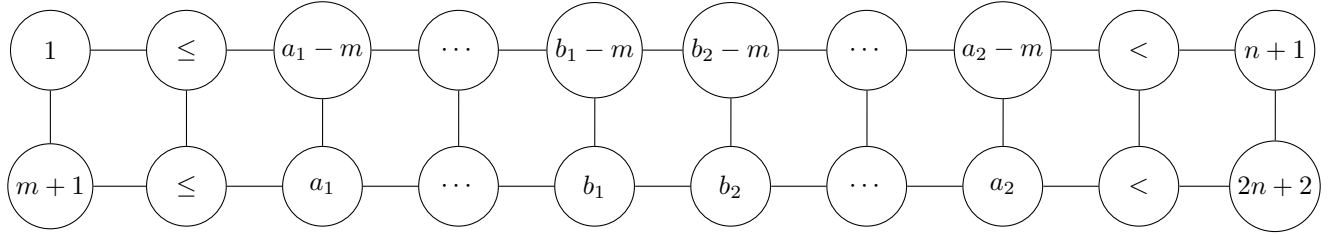
Figure 28



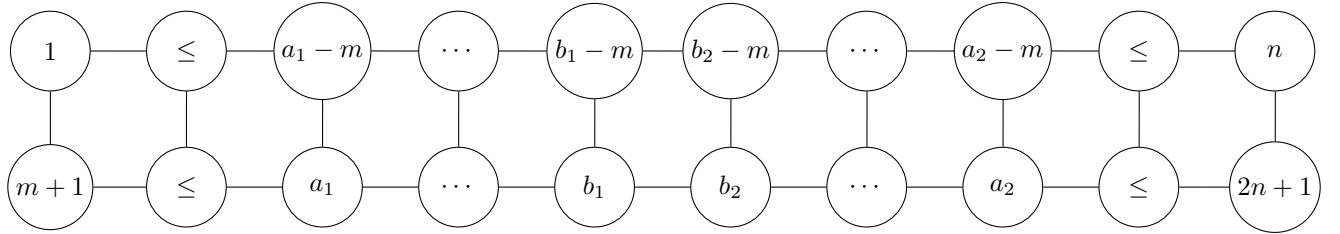
(a) Case 1: G is a $2 \times (n + 1)$ grid where $(n + 1) + 1 < a_1 < b_1 < b_2 < a_2 \leq 2(n + 1)$. Note: $m = n + 1$.



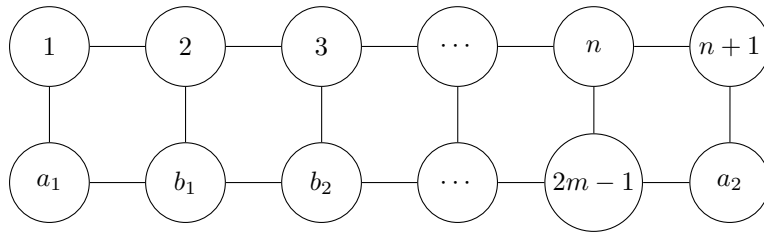
(b) Case 1: We can safely delete the first column and preserve all vertices of the bell pairs. The resulting grid is a $2 \times n$ grid, where $(n + 1) + 2 \leq a_1 < b_1 < b_2 < a_2 \leq 2(n + 1)$



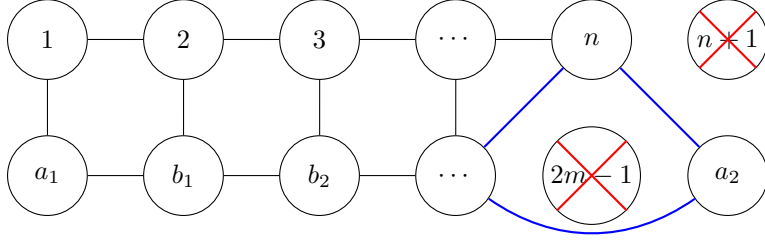
(c) Case 2: G is a $2 \times (n + 1)$ grid where $(n + 1) + 1 \leq a_1 < b_1 < b_2 < a_2 < 2(n + 1)$. Note: $m = n + 1$



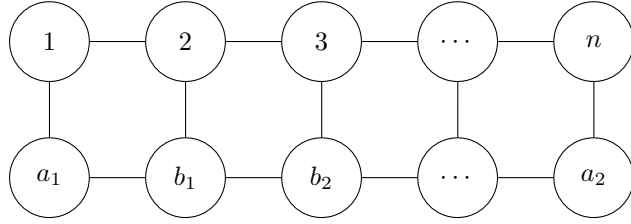
(d) Case 2: We can safely delete the last column and preserve all vertices of the bell pairs. The resulting grid is a $2 \times n$ grid, where $(n + 1) + 1 \leq a_1 < b_1 < b_2 < a_2 \leq 2n + 1$



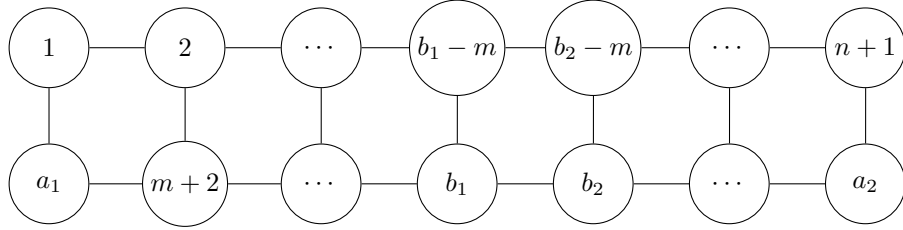
(e) Case 3: G is a $2 \times (n + 1)$ grid. $a_1 = m + 1$, $a_2 = 2m$ and $b_1 = a_1 + 1$. Then we know $b_2 = a_1 + 2$ and $b_2 \neq 2m - 1$. Note: $m = n + 1$.



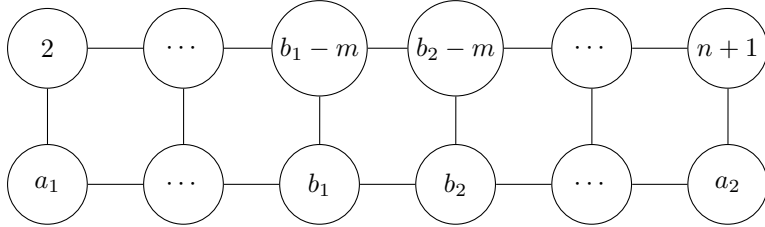
(f) Case 3: Delete the $\{n+1\}$ vertex and perform a Y measurement on $\{2m-1\}$. $G_1 = Y_{2m-1} \circ Z_{n+1}(G)$ Added edges from the LCs are colored.



(g) Case 3: We perform a local complementation on vertex $\{a_2\}$, so $G' = \tau_{a_2}(G_1)$. The resulting grid is a $2 \times n$ grid that preserves all vertices of the bell pairs.



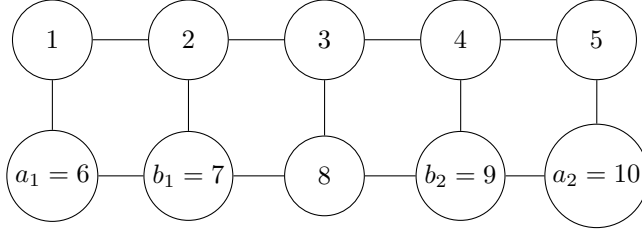
(h) Case 4: G is a $2 \times (n+1)$ grid. $a_1 = m+1$, $a_2 = 2m$ and $b_1 > a_1 + 1$. Then we know a_1 and b_1 cannot be neighbors. Note: $m = n+1$.



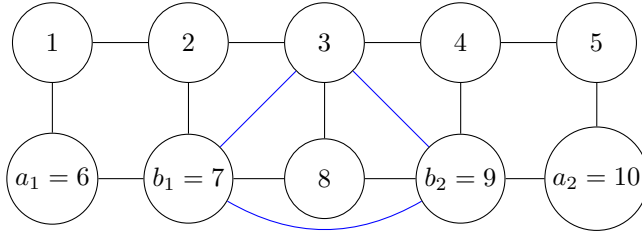
(i) Case 4: We perform a similar technique as in case 3, so $G' = \tau_{a_1} \circ Y_{m+2} \circ Z_1(G_1)$. The resulting grid is a $2 \times n$ grid that preserves all vertices of the bell pairs.

Figure 29

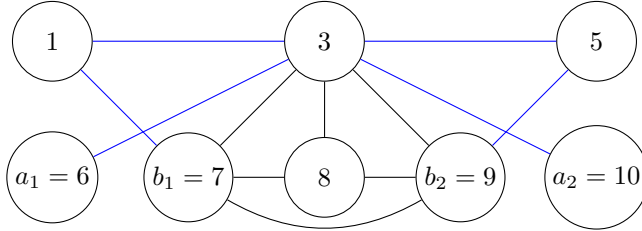
C.2 Intermediary Graphs for Lemma 3.17



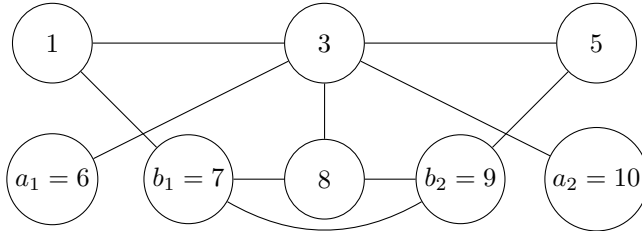
(a) G is a 2×5 grid. We will simultaneously extract $\{a_1, a_2\}$ $\{b_1, b_2\}$.



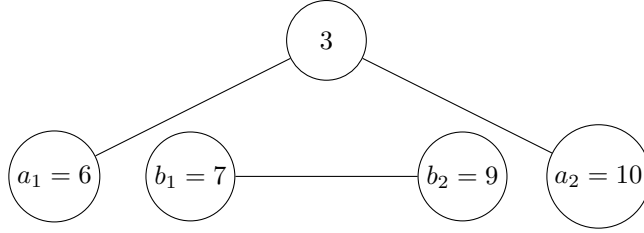
(b) G_1 , the first intermediary graph. Perform a local complementation on vertex $\{8\}$. $G_1 = \tau_8(G)$. Added edges from the LCs are colored.



(c) G_2 , the second intermediary graph. Perform X measurements on vertex $\{4\}$ using vertex $\{5\}$ as a pivot, and then on vertex $\{2\}$ using vertex $\{1\}$ as a pivot. $G_2 = X_2^1 \circ X_4^5(G_1)$. Added edges from the LCs are colored.



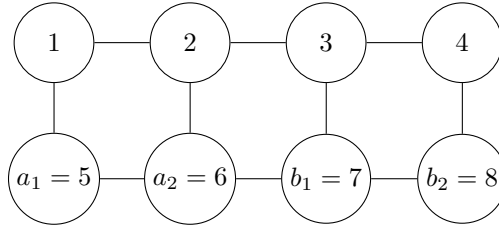
(d) G_3 , the third intermediary graph. Perform a local complementation on vertex $\{5\}$ and then on vertex $\{1\}$. $G_3 = \tau_1 \circ \tau_5(G_2)$.



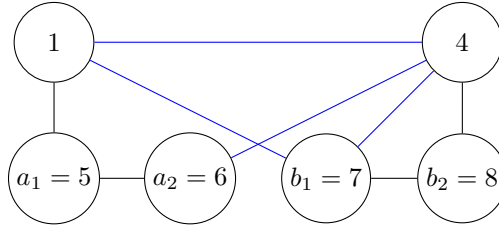
(e) G' , the resulting graph. Perform a Z measurement on vertex $\{1\}$, vertex $\{8\}$, and then on vertex $\{5\}$. $G' = Z_5 \circ Z_8 \circ Z_1(G_3) = Z_5 \circ Z_8 \circ Z_1 \circ \tau_1 \circ \tau_5 \circ X_2^1 \circ X_4^5 \circ \tau_8(G)$.

Figure 30

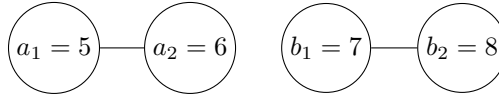
C.3 Intermediary Graphs for Lemma 3.18



(a) G' is a 2×4 grid. We will simultaneously extract $\{a_1, a_2\}$ $\{b_1, b_2\}$.



(b) G_1 , the first intermediary graph. Perform a Y measurement on vertex $\{2\}$ and then on vertex $\{3\}$. $G_1 = Y_3 \circ Y_2(G')$. Added edges from the LCs are colored.



(c) G'' , the resulting graph. Perform a Z measurement on vertex $\{1\}$ and then vertex $\{4\}$. $G'' = Z_4 \circ Z_1(G_1) = Z_4 \circ Z_1 \circ Y_3 \circ Y_2(G')$.

Figure 31